

Inverters

Converting dc to ac

8.1 INTRODUCTION

Inverters are circuits that convert dc to ac. More precisely, inverters transfer power from a dc source to an ac load. The controlled full-wave bridge converters in Chap. 4 can function as inverters in some instances, but an ac source must pre-exist in those cases. In other applications, the objective is to create an ac voltage when only a dc voltage source is available. The focus of this chapter is on inverters that produce an ac output from a dc input. Inverters are used in applications such as adjustable-speed ac motor drives, uninterruptible power supplies (UPS), and running ac appliances from an automobile battery.

8.2 THE FULL-BRIDGE CONVERTER

The full-bridge converter of Fig 8-1a is the basic circuit used to convert dc to ac. The full-bridge converter was introduced as part of a dc power supply circuit in Chap. 7. In this application, an ac output is synthesized from a dc input by closing and opening the switches in an appropriate sequence. The output voltage v_o can be $+V_{dc}$, $-V_{dc}$, or zero, depending on which switches are closed. Figure 8-1b to e shows the equivalent circuits for switch combinations.

Switches Closed	Output Voltage v_o
S_1 and S_2	$+V_{dc}$
S_3 and S_4	$-V_{dc}$
S_1 and S_3	0
S_2 and S_4	0

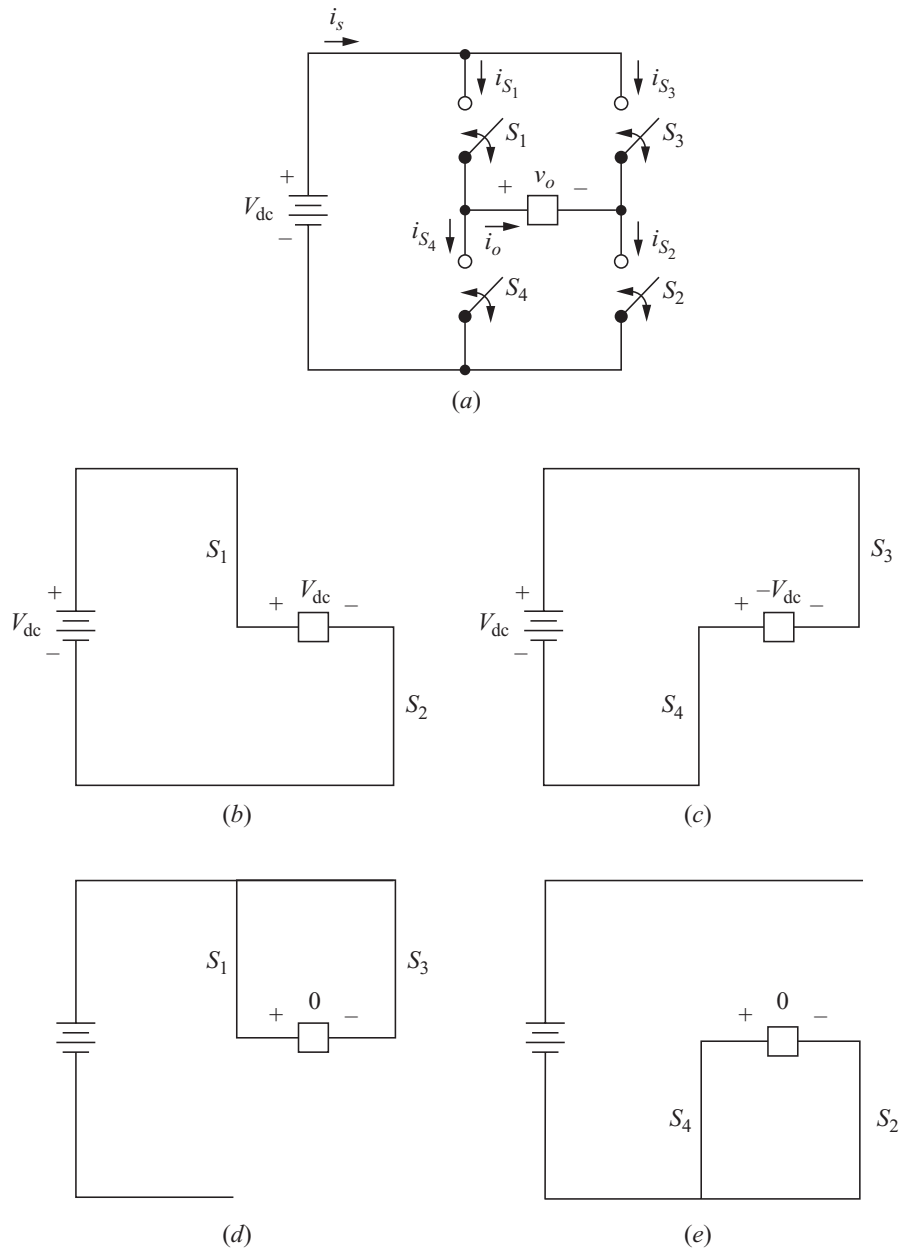


Figure 8-1 (a) Full-bridge converter; (b) S_1 and S_2 closed; (c) S_3 and S_4 closed; (d) S_1 and S_3 closed; (e) S_2 and S_4 closed.

Note that S_1 and S_4 should not be closed at the same time, nor should S_2 and S_3 . Otherwise, a short circuit would exist across the dc source. Real switches do not turn on or off instantaneously, as was discussed in Chap. 6. Therefore, switching transition times must be accommodated in the control of the switches. Overlap of switch “on” times will result in a short circuit, sometimes called a *shoot-through* fault, across the dc voltage source. The time allowed for switching is called *blanking* time.

8.3 THE SQUARE-WAVE INVERTER

The simplest switching scheme for the full-bridge converter produces a square wave output voltage. The switches connect the load to $+V_{dc}$ when S_1 and S_2 are closed or to $-V_{dc}$ when S_3 and S_4 are closed. The periodic switching of the load voltage between $+V_{dc}$ and $-V_{dc}$ produces a square wave voltage across the load. Although this alternating output is nonsinusoidal, it may be an adequate ac waveform for some applications.

The current waveform in the load depends on the load components. For the resistive load, the current waveform matches the shape of the output voltage. An inductive load will have a current that has more of a sinusoidal quality than the voltage because of the filtering property of the inductance. An inductive load presents some considerations in designing the switches in the full-bridge circuit because the switch currents must be bidirectional.

For a series RL load and a square wave output voltage, assume switches S_1 and S_2 in Fig. 8-1a close at $t = 0$. The voltage across the load is $+V_{dc}$, and current begins to increase in the load and in S_1 and S_2 . The current is expressed as the sum of the forced and natural responses

$$\begin{aligned} i_o(t) &= i_f(t) + i_n(t) \\ &= \frac{V_{dc}}{R} + Ae^{-t/\tau} \quad \text{for } 0 \leq t \leq T/2 \end{aligned} \quad (8-1)$$

where A is a constant evaluated from the initial condition and $\tau = L/R$.

At $t = T/2$, S_1 and S_2 open, and S_3 and S_4 close. The voltage across the RL load becomes $-V_{dc}$, and the current has the form

$$i_o(t) = \frac{-V_{dc}}{R} + Be^{-(t-T/2)/\tau} \quad \text{for } T/2 \leq t \leq T \quad (8-2)$$

where the constant B is evaluated from the initial condition.

When the circuit is first energized and the initial inductor current is zero, a transient occurs before the load current reaches a steady-state condition. At steady state, i_o is periodic and symmetric about zero, as illustrated in Fig. 8-2. Let the initial condition for the current described in Eq. (8-1) be $I_{\min}$, and let the initial condition for the current described in Eq. (8-2) be $I_{\max}$.

Evaluating Eq. (8-1) at $t = 0$,

$$i_o(0) = \frac{V_{dc}}{R} + Ae^0 = I_{\min}$$

or

$$A = I_{\min} - \frac{V_{dc}}{R} \quad (8-3)$$

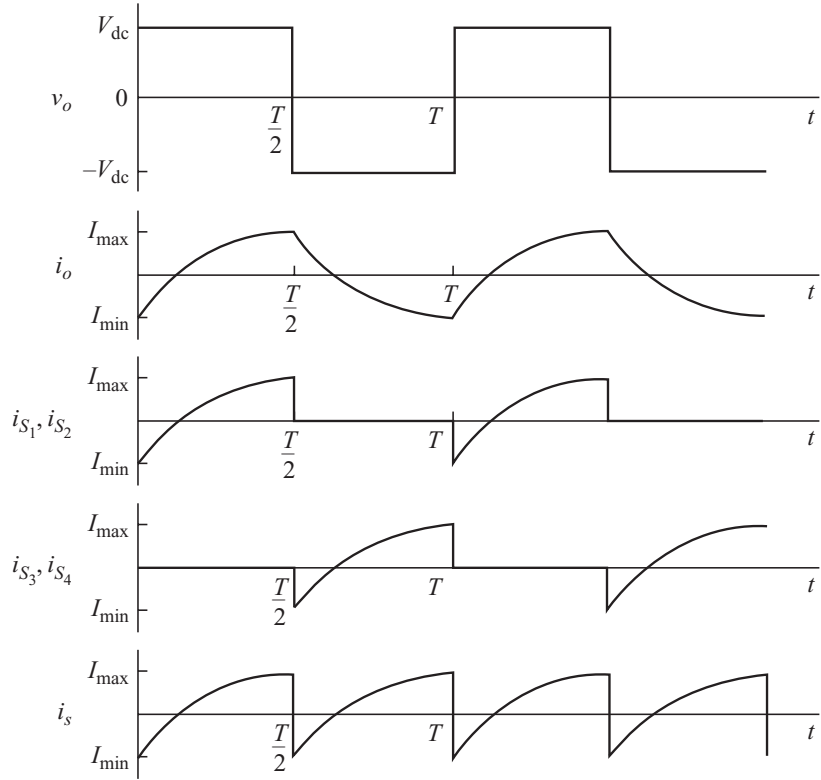


Figure 8-2 Square wave output voltage and steady-state current waveform for an RL load.

Likewise, Eq. (8-2) is evaluated at $t = T/2$.

$$i_o(T/2) = \frac{-V_{dc}}{R} + Be^0 = I_{max}$$

or

$$B = I_{max} + \frac{V_{dc}}{R} \quad (8-4)$$

In steady state, the current waveforms described by Eqs. (8-1) and (8-2) then become

$$i_o(t) = \begin{cases} \frac{V_{dc}}{R} + \left(I_{min} - \frac{V_{dc}}{R}\right)e^{-t/\tau} & \text{for } 0 < t < \frac{T}{2} \\ \frac{-V_{dc}}{R} + \left(I_{max} + \frac{V_{dc}}{R}\right)e^{-(t-T/2)/\tau} & \text{for } \frac{T}{2} < t < T \end{cases} \quad (8-5)$$

An expression is obtained for $I_{\max}$ by evaluating the first part of Eq. (8-5) at $t = T/2$

$$i(T/2) = I_{\max} = \frac{V_{\text{dc}}}{R} + \left(I_{\min} - \frac{V_{\text{dc}}}{R} \right) e^{-(T/2\tau)} \quad (8-6)$$

and by symmetry,

$$I_{\min} = -I_{\max} \quad (8-7)$$

Substituting $-I_{\max}$ for $I_{\min}$ in Eq. (8-6) and solving for $I_{\max}$,

$$I_{\max} = -I_{\min} = \frac{V_{\text{dc}}}{R} \left(\frac{1 - e^{-T/2\tau}}{1 + e^{-T/2\tau}} \right) \quad (8-8)$$

Thus, Eqs. (8-5) and (8-8) describe the current in an RL load in the steady state when a square wave voltage is applied. Figure 8-2 shows the resulting currents in the load, source, and switches.

Power absorbed by the load can be determined from $I_{\text{rms}}^2 R$, where rms load current is determined from the defining equation from Chap. 2. The integration may be simplified by taking advantage of the symmetry of the waveform. Since the square each of the current half-periods is identical, only the first half-period needs to be evaluated:

$$I_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt} = \sqrt{\frac{2}{T} \int_0^{T/2} \left[\frac{V_{\text{dc}}}{R} + \left(I_{\min} - \frac{V_{\text{dc}}}{R} \right) e^{-t/\tau} \right]^2 dt} \quad (8-9)$$

If the switches are ideal, the power supplied by the source must be the same as absorbed by the load. Power from a dc source is determined from

$$P_{\text{dc}} = V_{\text{dc}} I_s \quad (8-10)$$

as was derived in Chap. 2.

EXAMPLE 8-1

Square-Wave Inverter with RL Load

The full-bridge inverter of Fig. 8-1 has a switching sequence that produces a square wave voltage across a series RL load. The switching frequency is 60 Hz, $V_{\text{dc}} = 100$ V, $R = 10 \Omega$, and $L = 25$ mH. Determine (a) an expression for load current, (b) the power absorbed by the load, and (c) the average current in the dc source.

■ Solution

(a) From the parameters given,

$$T = 1/f = 1/60 = 0.0167 \text{ s}$$

$$\tau = L/R = 0.025/10 = 0.0025 \text{ s}$$

$$T/2\tau = 3.33$$

Equation (8-8) is used to determine the maximum and minimum current.

$$I_{\max} = -I_{\min} = \frac{100}{10} \left(\frac{1 - e^{-3.33}}{1 + e^{-3.33}} \right) = 9.31 \text{ A}$$

Equation (8-5) is then evaluated to give load current.

$$\begin{aligned} i_o(t) &= \frac{100}{10} + \left(-9.31 - \frac{100}{10} \right) e^{-t/0.0025} \\ &= 10 - 19.31 e^{-t/0.0025} \quad 0 \leq t \leq \frac{1}{120} \\ i_o(t) &= -\frac{100}{10} + \left(9.31 + \frac{100}{10} \right) e^{-(t-0.0167/2)/0.0025} \\ &= -10 + 19.31 e^{-(t-0.00835)/0.0025} \quad \frac{1}{120} \leq t \leq \frac{1}{60} \end{aligned}$$

(b) Power is computed from $I_{\text{rms}}^2 R$, where I_{rms} is computed from Eq. (8-9).

$$I_{\text{rms}} = \sqrt{\frac{1}{120} \int_0^{1/120} [(10 - 19.31) e^{-t/0.0025}]^2 dt} = 6.64 \text{ A}$$

Power absorbed by the load is

$$P = I_{\text{rms}}^2 R = (6.64)^2 (10) = 441 \text{ W}$$

(c) Average source current can also be computed by equating source and load power, assuming a lossless converter. Using Eq. (8-10),

$$I_s = \frac{P_{\text{dc}}}{V_{\text{dc}}} = \frac{441}{100} = 4.41 \text{ A}$$

Average power could also be computed from the average of the current expression in part (a).

The switch currents in Fig. 8-2 show that the switches in the full-bridge circuit must be capable of carrying both positive and negative currents for RL loads. However, real electronic devices may conduct current in one direction only. This problem is solved by placing feedback diodes in parallel (antiparallel) with each switch. During the time interval when the current in the switch must be negative, the feedback diode carries the current. The diodes are reverse-biased when current is positive in the switch. Figure 8-3a shows the full-bridge inverter with switches implemented as insulated gate bipolar transistors (IGBTs) with feedback diodes. Transistor and diode currents for a square wave voltage and an RL load are indicated in Fig 8-3b. Power semiconductor modules often include feedback diodes with the switches.

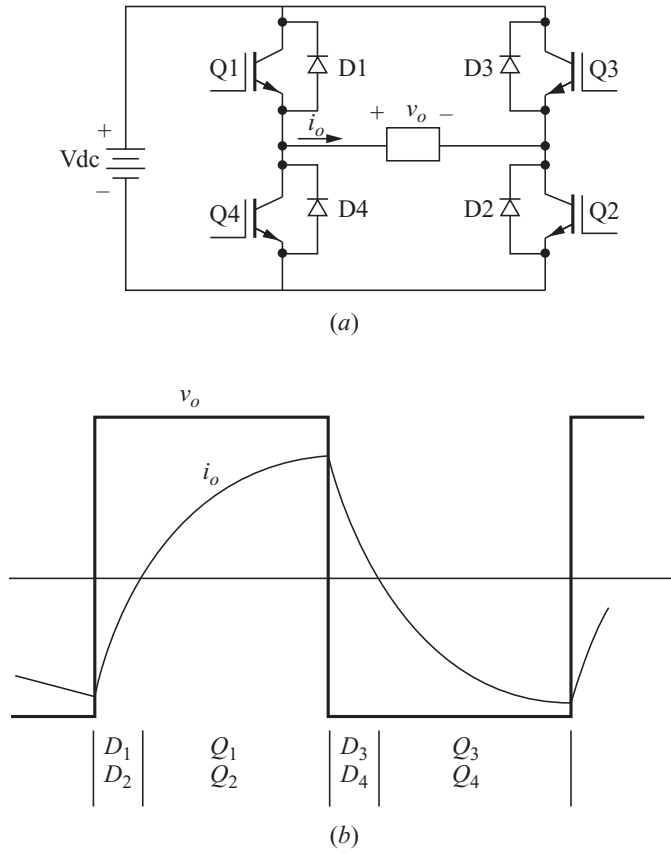


Figure 8-3 (a) Full-bridge inverter using IGBTs;
(b) Steady-state current for an RL load.

When IGBTs Q_1 and Q_2 are turned off in Fig. 8-3a, the load current must be continuous and will transfer to diodes D_3 and D_4 , making the output voltage $-V_{dc}$, effectively turning on the switch paths 3 and 4 before Q_3 and Q_4 are turned on. IGBTs Q_3 and Q_4 must be turned on before the load current decays to zero.

8.4 FOURIER SERIES ANALYSIS

The Fourier series method is often the most practical way to analyze load current and to compute power absorbed in a load, especially when the load is more complex than a simple resistive or RL load. A useful approach for inverter analysis is to express the output voltage and load current in terms of a Fourier series. With no dc component in the output,

$$v_o(t) = \sum_{n=1}^{\infty} V_n \sin(n\omega_0 t + \theta_n) \quad (8-11)$$

and

$$i_o(t) = \sum_{n=1}^{\infty} I_n \sin(n\omega_0 t + \phi_n) \quad (8-12)$$

Power absorbed by a load with a series resistance is determined from $I_{\text{rms}}^2 R$, where the rms current can be determined from the rms currents at each of the components in the Fourier series by

$$I_{\text{rms}} = \sqrt{\sum_{n=1}^{\infty} I_{n,\text{rms}}^2} = \sqrt{\sum_{n=1}^{\infty} \left(\frac{I_n}{\sqrt{2}} \right)^2} \quad (8-13)$$

where

$$I_n = \frac{V_n}{Z_n} \quad (8-14)$$

and Z_n is the load impedance at harmonic n .

Equivalently, the power absorbed in the load resistor can be determined for each frequency in the Fourier series. Total power can be determined from

$$P = \sum_{n=1}^{\infty} P_n = \sum_{n=1}^{\infty} I_{n,\text{rms}}^2 R \quad (8-15)$$

where $I_{n,\text{rms}}$ is $I_n/\sqrt{2}$.

In the case of the square wave, the Fourier series contains the odd harmonics and can be represented as

$$v_o(t) = \sum_{n \text{ odd}} \frac{4V_{\text{dc}}}{n\pi} \sin n\omega_0 t \quad (8-16)$$

EXAMPLE 8-2

Fourier Series Solution for the Square-Wave Inverter

For the inverter in Example 8-1 ($V_{\text{dc}} = 100 \text{ V}$, $R = 10 \Omega$, $L = 25 \text{ mH}$, $f = 60 \text{ Hz}$), determine the amplitudes of the Fourier series terms for the square wave load voltage, the amplitudes of the Fourier series terms for load current, and the power absorbed by the load.

■ Solution

The load voltage is represented as the Fourier series in Eq. (8-16). The amplitude of each voltage term is

$$V_n = \frac{4V_{\text{dc}}}{n\pi} = \frac{4(100)}{n\pi}$$

The amplitude of each current term is determined from Eq. (8-14),

$$I_n = \frac{V_n}{Z_n} = \frac{V_n}{\sqrt{R^2 + (n\omega_0 L)^2}} = \frac{4(100)/n\pi}{\sqrt{10^2 + [n(2\pi 60)(0.025)]^2}}$$

Power at each frequency is determined from Eq. (8-15).

$$P_n = I_{n,\text{rms}}^2 R = \left(\frac{I_n}{\sqrt{2}} \right)^2 R$$

Table 8-1 Fourier Series Quantities for Example 8-2

n	f_n (Hz)	V_n (V)	Z_n (Ω)	I_n (A)	P_n (W)
1	60	127.3	13.7	9.27	429.3
3	180	42.4	30.0	1.42	10.0
5	300	25.5	48.2	0.53	1.40
7	420	18.2	66.7	0.27	0.37
9	540	14.1	85.4	0.17	0.14

Table 8-1 summarizes the Fourier series quantities for the circuit of Example 8-1. As the harmonic number n increases, the amplitude of the Fourier voltage component decreases and the magnitude of the corresponding impedance increases, both resulting in small currents for higher-order harmonics. Therefore, only the first few terms of the series are of practical interest. Note how the current and power terms become vanishingly small for all but the first few frequencies.

Power absorbed by the load is computed from Eq. (8-15).

$$P = \sum P_n = 429.3 + 10.0 + 1.40 + 0.37 + 0.14 + \cdots \approx 441 \text{ W}$$

which agrees with the result in Example 8-1.

8.5 TOTAL HARMONIC DISTORTION

Since the objective of the inverter is to use a dc voltage source to supply a load requiring ac, it is useful to describe the quality of the ac output voltage or current. The quality of a nonsinusoidal wave can be expressed in terms of total harmonic distortion (THD), defined in Chap. 2. Assuming no dc component in the output,

$$\text{THD} = \frac{\sqrt{\sum_{n=2}^{\infty} (V_{n,\text{rms}})^2}}{V_{1,\text{rms}}} = \frac{\sqrt{V_{\text{rms}}^2 - V_{1,\text{rms}}^2}}{V_{1,\text{rms}}} \quad (8-17)$$

The THD of current is determined by substituting current for voltage in the above equation. The THD of load current is often of greater interest than that of output voltage. This definition for THD is based on the Fourier series, so there is some benefit in using the Fourier series method for analysis when the THD must be determined. Other measures of distortion such as distortion factor, as presented in Chap. 2, can also be applied to describe the output waveform for inverters.

EXAMPLE 8-3

THD for a Square-Wave Inverter

Determine the total harmonic distortion of the load voltage and the load current for the square-wave inverter in Examples 8-1 and 8-2.

■ Solution

Use the Fourier series for the square wave in Eq. (8-16) and the definition of THD in Eq. (8-17). The rms value of the square wave voltage is the same as the peak value, and the fundamental frequency component is the first term in Eq. (8-16),

$$V_{\text{rms}} = V_{\text{dc}}$$

$$V_{1,\text{rms}} = \frac{V_1}{\sqrt{2}} = \frac{4V_{\text{dc}}}{\sqrt{2}\pi}$$

Using Eq. (8-17) to compute the total harmonic distortion for voltage,

$$\text{THD}_V = \frac{\sqrt{V_{\text{rms}}^2 - V_{1,\text{rms}}^2}}{V_{1,\text{rms}}} = \frac{\sqrt{V_{\text{dc}}^2 - (4V_{\text{dc}}/\sqrt{2}\pi)^2}}{4V_{\text{dc}}/\sqrt{2}\pi} = 0.483 = 48.3\%$$

The THD of the current is computed using the truncated Fourier series which was determined in Example 8-2.

$$\begin{aligned} \text{THD}_I &= \frac{\sqrt{\sum_{n=2}^{\infty} (I_{n,\text{rms}})^2}}{I_{1,\text{rms}}} \\ &\approx \frac{\sqrt{(1.42/\sqrt{2})^2 + (0.53/\sqrt{2})^2 + (0.27/\sqrt{2})^2 + (0.17/\sqrt{2})^2}}{9.27/\sqrt{2}} = 0.167 = 16.7\% \end{aligned}$$

8.6 PSPICE SIMULATION OF SQUARE-WAVE INVERTERS

Computer simulation of inverter circuits can include various levels of circuit detail. If only the current waveform in the load is desired, it is sufficient to provide a source that will produce the appropriate voltage that would be expected on the inverter output. For example, a full-bridge inverter producing a square wave output might be replaced with a square wave voltage source using the VPULSE source. This simplified simulation will predict the behavior of the current in the load but will give no direct information about the switches. Also, this approach assumes that the switching operation correctly produces the desired output.

EXAMPLE 8-4

PSpice Simulation for Example 8-1

For a series (RL) load in a full-bridge inverter circuit with a square wave output, the dc supply is 100 V, $R = 10 \Omega$, $L = 25 \text{ mH}$, and the switching frequency is 60 Hz (Example 8-1). (a) Assuming ideal switches, use PSpice to determine the maximum and minimum

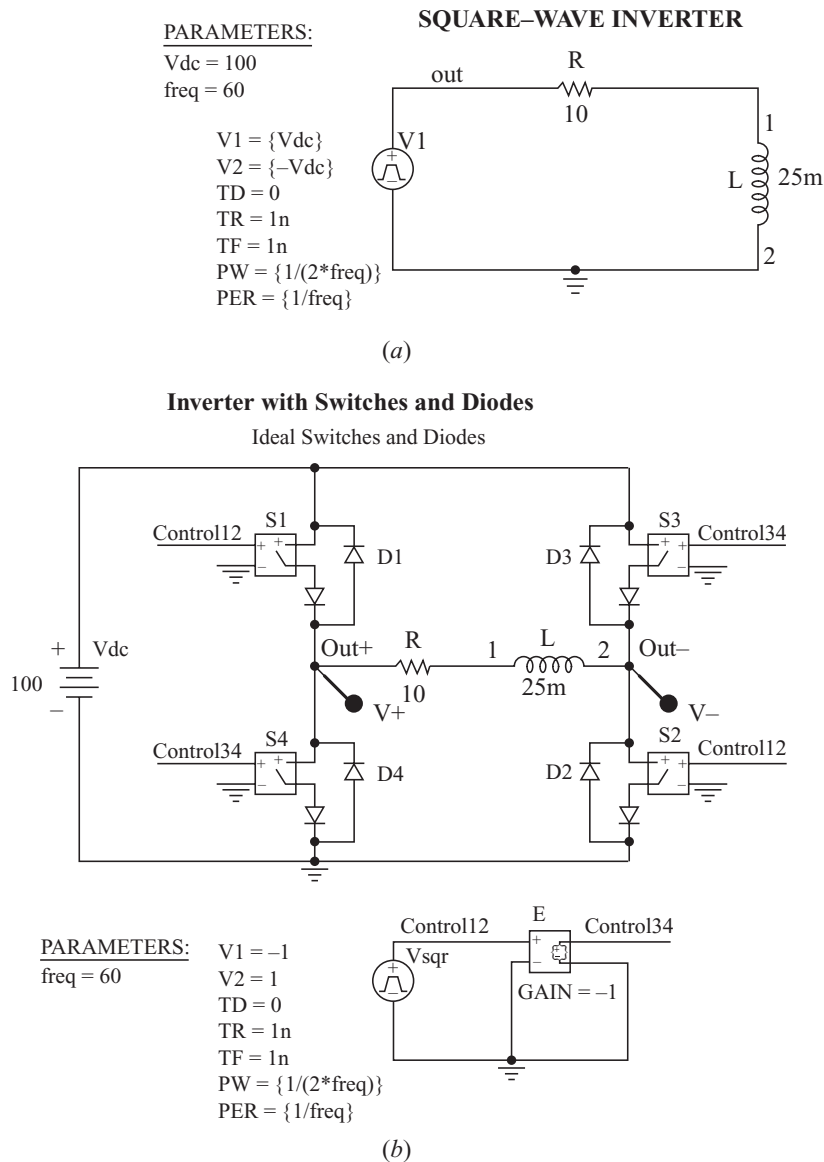


Figure 8-4 (a) Square-wave inverter simulation using an ideal source;
 (b) Square-wave inverter using switches and diodes.

current in the load in the steady state. (b) Determine the power absorbed by the load.
 (c) Determine the total harmonic distortion of the load current.

■ Solution 1

Since individual switch currents are not of concern in this problem, a square wave voltage source (VPULSE), as shown in Fig. 8-4a, across the load can simulate the converter output.

Set up a simulation profile for a transient analysis having a run time of 50 ms (three periods), and start saving data after 16.67 ms (one period) so the output represents steady-state current.

Fourier analysis is performed under *Simulation Settings, Output File Options, Perform Fourier Analysis, Center Frequency: 60 Hz, Number of Harmonics: 15, Output Variables: V(OUT) I(R)*.

- (a) When in Probe, enter the expression $I(R)$ to obtain a display of the current in the load resistor. The first period contains the start-up transient, but steady-state current like that in Fig. 8-2 is displayed thereafter. The maximum and minimum steady-state current values are approximately 9.31 and -9.31 A, which can be obtained precisely by using the cursor option.
- (b) Average power can be obtained from Probe by displaying load current, making sure that the data represent the steady-state condition and entering the expression $\text{AVG}(W(R))$ or $\text{AVG}(V(\text{OUT}) \cdot I(R))$. This shows that the resistor absorbs approximately 441 W. The rms current is determined by entering $\text{RMS}(I(R))$, resulting in 6.64 A, as read from the end of the trace. These results agree with the analysis in Example 8-1.
- (c) The THD is obtained from the Fourier series for $I(R)$ in the output file as 16.7 percent, agreeing with the Fourier analysis in Examples 8-2 and 8-3. Note that the THD for the square wave in the output file is 45 percent, which is lower than the 48.3 percent computed in Example 8-3. The THD in PSpice is based on the truncated Fourier series through $n = 15$. The magnitudes of higher-order harmonics are not insignificant for the square wave, and omitting them underestimates the THD. The higher-order current harmonics are small, so there is little error in omitting them from the analysis. The number of harmonics in the output file can be increased if desired.

■ Solution 2

The inverter is simulated using the full-bridge circuit of Fig. 8-4b. (This requires the full version of PSpice.) The result of this simulation gives information about the currents and voltages for the switching devices. Voltage-controlled switches (Sbreak) and the default diode (Dbreak) are used. Diodes are included in the switch model to make the switches unidirectional. The model for Sbreak is changed so $R_{\text{on}} = 0.01 \, \Omega$, and the model for Dbreak is changed so $n = 0.01$, approximating an ideal diode. The output voltage is between nodes out+ and out-. Models for the switches and diodes can be changed to determine the behavior of the circuit using realistic switching devices.

8.7 AMPLITUDE AND HARMONIC CONTROL

The amplitude of the fundamental frequency for a square wave output from the full-bridge inverter is determined by the dc input voltage [Eq. (8-16)]. A controlled output can be produced by modifying the switching scheme. An output voltage of the form shown in Fig. 8-5a has intervals when the output is zero as well as $+V_{\text{dc}}$ and $-V_{\text{dc}}$. This output voltage can be controlled by adjusting the interval α on each side of the pulse where the output is zero. The rms value of the voltage waveform in Fig 8-5a is

$$V_{\text{rms}} = \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi-\alpha} V_{\text{dc}}^2 d(\omega t)} = V_{\text{dc}} \sqrt{1 - \frac{2\alpha}{\pi}} \quad (8-18)$$

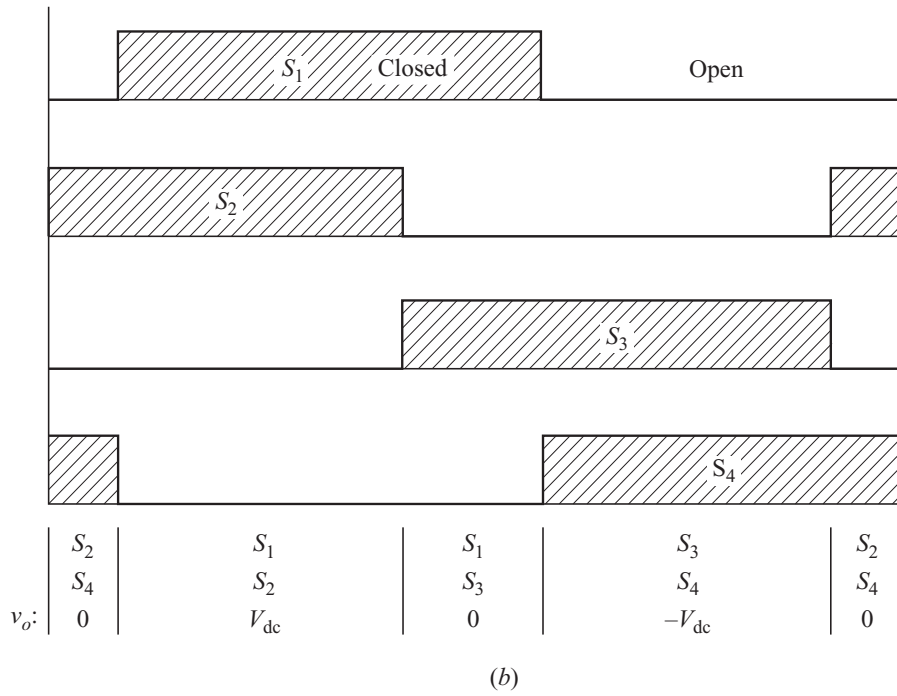
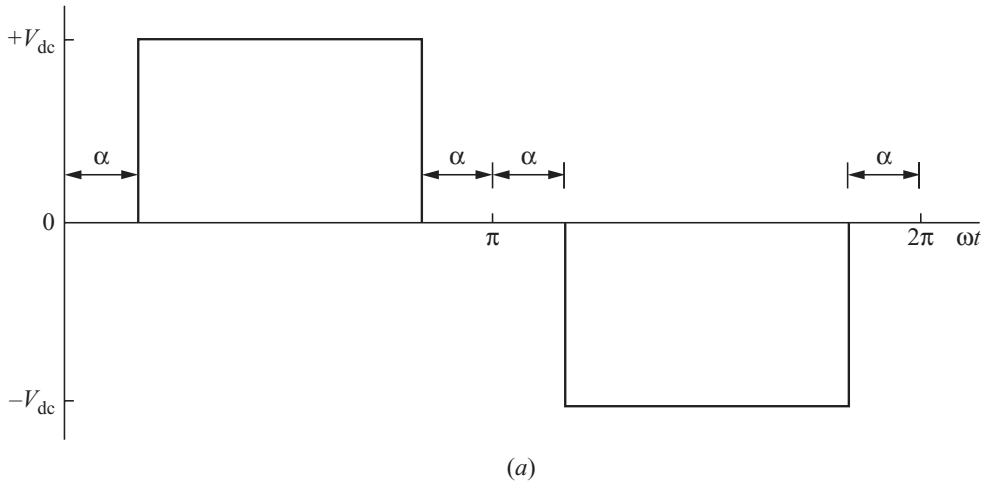


Figure 8-5 (a) Inverter output for amplitude and harmonic control; (b) Switching sequence for the full-bridge inverter of Fig. 8-1a.

The Fourier series of the waveform is expressed as

$$v_o(t) = \sum_{n \text{ odd}} V_n \sin(n\omega_0 t) \quad (8-19)$$

Taking advantage of half-wave symmetry, the amplitudes are

$$V_n = \frac{2}{\pi} \int_{\alpha}^{\pi - \alpha} V_{dc} \sin(n\omega_0 t) d(\omega_0 t) = \frac{4V_{dc}}{n\pi} \cos(n\alpha) \quad (8-20)$$

where α is the angle of zero voltage on each end of the pulse. The amplitude of each frequency of the output is a function of α . In particular, the amplitude of the fundamental frequency ($n = 1$) is controllable by adjusting α :

$$V_1 = \left(\frac{4V_{dc}}{\pi} \right) \cos \alpha \quad (8-21)$$

Harmonic content can also be controlled by adjusting α . If $\alpha = 30^\circ$, for example, $V_3 = 0$. This is significant because the third harmonic can be eliminated from the output voltage and current. Other harmonics can be eliminated by choosing a value of α which makes the cosine term in Eq. (8-20) to go to zero. Harmonic n is eliminated if

$$\alpha = \frac{90^\circ}{n} \quad (8-22)$$

The switching scheme required to produce an output like Fig. 8-5a must provide intervals when the output voltage is zero, as well as $\pm V_{dc}$. The switching sequence of Fig. 8-5b is a way to implement the required output waveform.

Amplitude control and harmonic reduction may not be compatible. For example, establishing α at 30° to eliminate the third harmonic fixes the amplitude of the output fundamental frequency at $V_1 = (4V_{dc}/\pi) \cos 30^\circ = 1.1 V_{dc}$ and removes further controllability. To control both amplitude and harmonics using this switching scheme, it is necessary to be able to control the dc input voltage to the inverter. A dc-dc converter (Chap. 6 and 7) placed between the dc source and the inverter can provide a controlled dc input to the inverter.

A graphical representation of the integration in the Fourier series coefficient of Eq. (8-20) gives some insight into harmonic elimination. Recall from Chap. 2 that the Fourier coefficients are determined from the integral of the product of the waveform and a sinusoid. Figure 8-6a shows the output waveform for $\alpha = 30^\circ$ and the sinusoid of $\omega = 3\omega_o$. The product of these two waveforms has an area of zero, showing that the third harmonic is zero. Figure 8-6b shows the waveform for $\alpha = 18^\circ$ and the sinusoid of $\omega = 5\omega_o$, showing that the fifth harmonic is eliminated for this value of α .

Other switching schemes can eliminate multiple harmonics. For example, the output waveform shown in Fig. 8-6c eliminates both the third and fifth harmonics, as indicated by the areas of both being zero.

EXAMPLE 8-5

Harmonic Control of the Full-Bridge Inverter Output

Design an inverter that will supply the series RL load of the previous examples ($R = 10 \Omega$ and $L = 25 \text{ mH}$) with a fundamental-frequency current amplitude of 9.27 A, but with a THD of less than 10 percent. A variable dc source is available.

■ Solution

A square-wave inverter produces a THD for current of 16.7 percent (Example 8-3), which does not meet the specification. The dominant harmonic current is for $n = 3$, so a switching

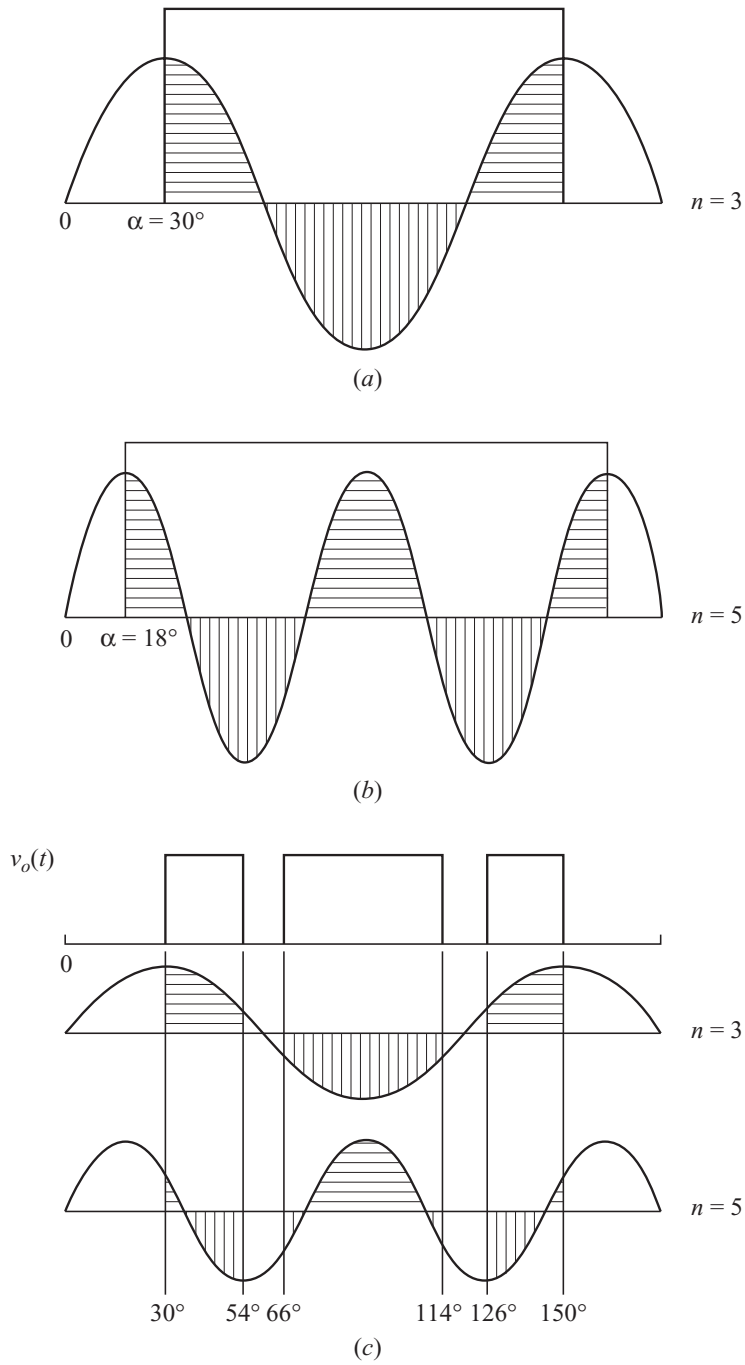


Figure 8-6 Harmonic elimination; (a) Third harmonic; (b) Fifth harmonic; (c) Third and fifth harmonics.

scheme to eliminate the third harmonic will reduce the THD. The required voltage amplitude at the fundamental frequency is

$$V_1 = I_1 Z_1 = I_1 \sqrt{R^2 + (\omega_0 L)^2} = 9.27 \sqrt{10^2 + [2\pi 60 (0.025)]^2} = 127 \text{ V}$$

Using the switching scheme of Fig. 8-5b, Eq. (8-21) describes the amplitude of the fundamental-frequency voltage,

$$V_1 = \left(\frac{4V_{dc}}{\pi} \right) \cos \alpha$$

Solving for the required dc input with $\alpha = 30^\circ$,

$$V_{dc} = \frac{V_1 \pi}{4 \cos \alpha} = \frac{127\pi}{4 \cos 30^\circ} = 116 \text{ V}$$

Other harmonic voltages are described by Eq. (8-20), and currents for these harmonics are determined from voltage amplitude and load impedance using the same technique as for the square-wave inverter of Example 8-2. The results are summarized in Table 8-2.

Table 8-2 Fourier Series Quantities for Example 8-5

n	f_n (Hz)	V_n (V)	Z_n (Ω)	I_n (A)
1	60	127	13.7	9.27
3	180	0	30.0	0
5	300	25.5	48.2	0.53
7	420	18.2	66.7	0.27
9	540	0	85.4	0
11	660	11.6	104	0.11

The THD of the load current is then

$$\text{THD}_1 = \frac{\sqrt{\sum_{n=2}^{\infty} I_{n,\text{rms}}^2}}{I_{1,\text{rms}}} \approx \frac{\sqrt{(0.53/\sqrt{2})^2 + (0.27/\sqrt{2})^2 + (0.11/\sqrt{2})^2}}{9.27/\sqrt{2}} = 0.066 = 6.6\%$$

which more than satisfies the design specifications.

A PSpice circuit for the full-bridge inverter with harmonic and amplitude control is shown in Fig. 8-7a. The user must enter the parameters alpha, output fundamental frequency, dc input voltage to the bridge, and load. The Probe output for voltage and current is shown in Fig. 8-7b. The current is scaled by a factor of 10 to show its relationship to the voltage waveform. The THD of the load current is obtained from the Fourier analysis in the output file as 6.6 percent.

8.8 THE HALF-BRIDGE INVERTER

The half-bridge converter of Fig. 8-8 can be used as an inverter. This circuit was introduced in Chap. 7 as applied to dc power supply circuits. In this circuit, the number of switches is reduced to 2 by dividing the dc source voltage into two parts with the capacitors. Each capacitor will be the same value and will have

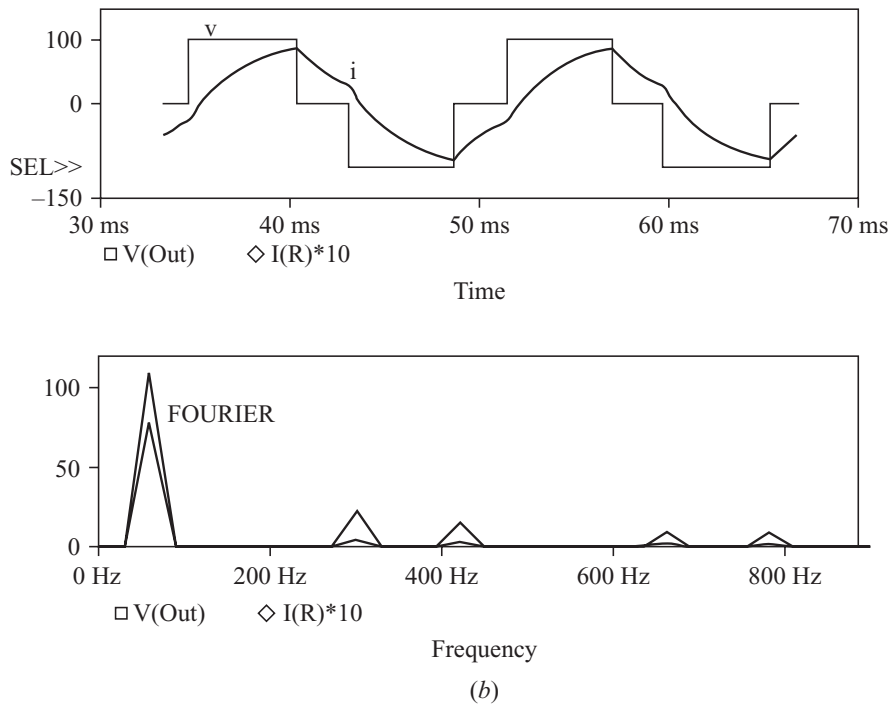
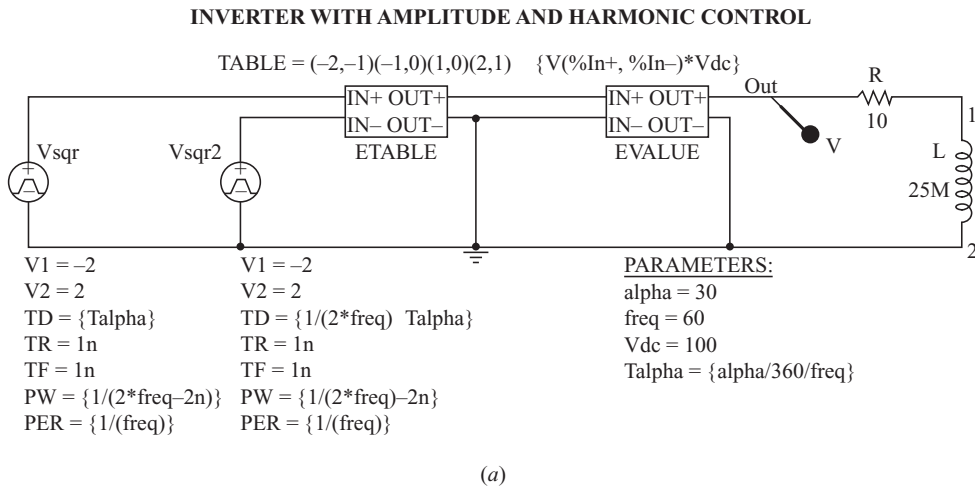


Figure 8-7 (a) A PSpice circuit for Example 8-5 to produce the voltage waveform in Fig. 8-5a; (b) Probe output for showing harmonic elimination.

voltage $V_{dc}/2$ across it. When S_1 is closed, the load voltage is $-V_{dc}/2$. When S_2 is closed, the load voltage is $+V_{dc}/2$. Thus, a square wave output or a bipolar pulse-width-modulated output, as described in Sec. 8.10, can be produced.

The voltage across an open switch is twice the load voltage, or V_{dc} . As with the full-bridge inverter, blanking time for the switches is required to prevent a short circuit across the source, and feedback diodes are required to provide continuity of current for inductive loads.

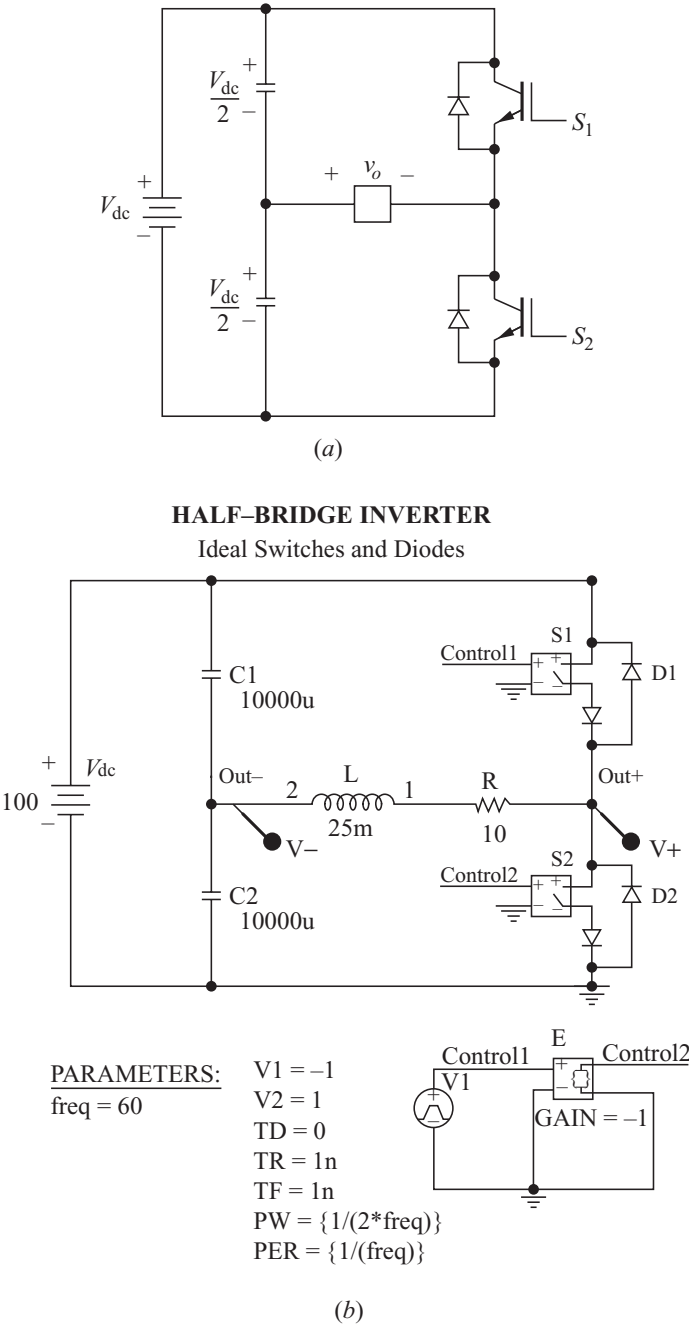


Figure 8-8 (a) A half-bridge inverter using IGBTs. The output is $\pm V_{dc}$; (b) A PSpice implementation using voltage-controlled switches and diodes.

8.9 MULTILEVEL INVERTERS

The H bridge inverter previously illustrated in Figs. 8-1 and 8-3 produces output voltages of V_{dc} , 0, and $-V_{dc}$. The basic H bridge switching concept can be expanded to other circuits that can produce additional output voltage levels.

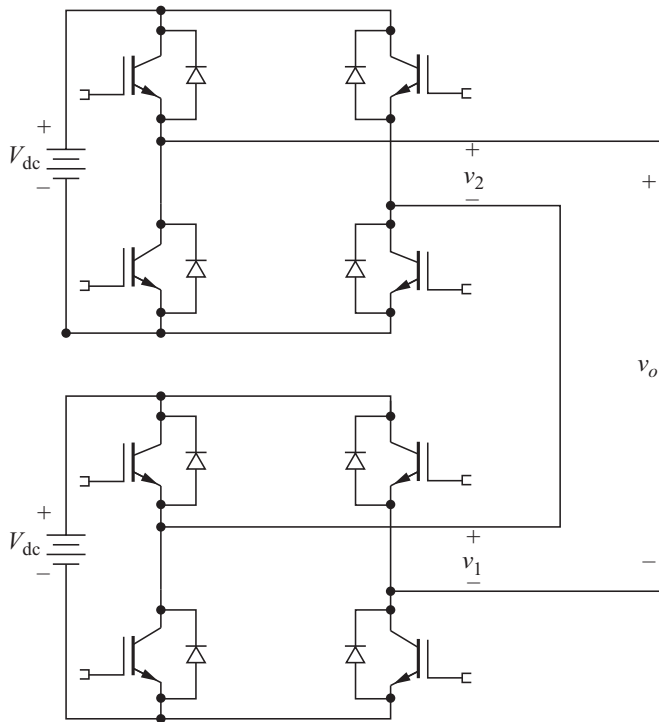


Figure 8-9 An inverter with two dc sources, each with an H bridge implemented with IGBTs.

These multilevel-output voltages are more sinelike in quality and thus reduce harmonic content. The multilevel inverter is suitable for applications including adjustable-speed motor drives and interfacing renewable energy sources such as photovoltaics to the electric power grid.

Multilevel Converters with Independent DC Sources

One multilevel inverter method uses independent dc sources, each with an H bridge. A circuit with two dc voltage sources is shown in Fig. 8-9. The output of each of the H bridges is $+V_{dc}$, $-V_{dc}$, or 0, as was illustrated in Fig. 8-1. The total instantaneous voltage v_o on the output of the multilevel converter is any combination of individual bridge voltages. Thus, for a two-source inverter, v_o can be any of the five levels $+2V_{dc}$, V_{dc} , 0, $-V_{dc}$, or $-2V_{dc}$.

Each H bridge operates with a switching scheme like that of Fig. 8.5 in Sec. 8.7, which was used for amplitude or harmonic control. Each bridge operates at a different delay angle α , resulting in bridge and total output voltages like those shown in Fig. 8-10.

The Fourier series for the total output voltage v_o for the two-source circuit contains only the odd-numbered harmonics and is

$$v_o(t) = \frac{4V_{dc}}{\pi} \sum_{n=1,3,5,7,\dots}^{\infty} [\cos(n\alpha_1) + \cos(n\alpha_2)] \frac{\sin(n\omega_0 t)}{n} \quad (8-23)$$

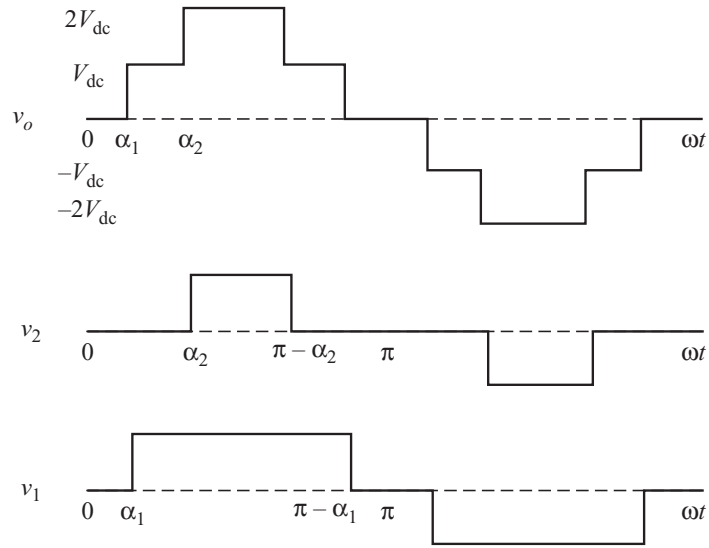


Figure 8-10 Voltage output of each of the H bridges and the total voltage for the two-source multilevel inverter of Fig. 8-9.

The Fourier coefficients for this series are

$$V_n = \frac{4V_{dc}}{n\pi} [\cos(n\alpha_1) + \cos(n\alpha_2)] \quad (8-24)$$

The modulation index M_i is the ratio of the amplitude of the fundamental frequency component of v_o to the amplitude of the fundamental frequency component of a square wave of amplitude $2V_{dc}$, which is $2(4V_{dc}/\pi)$.

$$M_i = \frac{V_1}{2(4V_{dc}/\pi)} = \frac{\cos \alpha_1 + \cos \alpha_2}{2} \quad (8-25)$$

Some harmonics can be eliminated from the output voltage waveform with the proper selection of α_1 and α_2 in Eq. (8-24). For the two-source converter, harmonic m can be eliminated by using delay angles such that

$$\cos(m\alpha_1) + \cos(m\alpha_2) = 0 \quad (8-26)$$

To eliminate the m th harmonic and also meet a specified modulation index for the two-source inverter requires the simultaneous solution to Eq. (8-26) and the additional equation derived from Eq. (8-25),

$$\cos(\alpha_1) + \cos(\alpha_2) = 2M_i \quad (8-27)$$

To solve Eqs. (8-26) and (8-27) simultaneously requires an iterative numerical method such as the Newton-Raphson method.

EXAMPLE 8-6

A Two-Source Multilevel Inverter

For the two-source multilevel inverter of Fig. 8-9 with $V_{dc} = 100$ V: (a) Determine the Fourier coefficients through $n = 9$ and the modulation index for $\alpha_1 = 20^\circ$ and $\alpha_2 = 40^\circ$. (b) Determine α_1 and α_2 such that the third harmonic ($n = 3$) is eliminated and $M_i = 0.8$.

■ Solution

Using Eq. 8-24 to evaluate the Fourier coefficients,

$$V_n = \frac{4V_{dc}}{n\pi} [\cos(n\alpha_1) + \cos(n\alpha_2)] = \frac{4(100)}{n\pi} [\cos(n20^\circ) + \cos(n40^\circ)]$$

resulting in $V_1 = 217$, $V_3 = 0$, $V_5 = -28.4$, $V_7 = -10.8$, and $V_9 = 0$. Note that the third and ninth harmonics are eliminated. The even harmonics are not present.

The modulation index M_i is evaluated from Eq. (8-25).

$$M_i = \frac{\cos\alpha_1 + \cos\alpha_2}{2} = \frac{\cos 20^\circ + \cos 40^\circ}{2} = 0.853$$

The amplitude of the fundamental frequency voltage is therefore 85.3 percent of that of a square wave of ± 200 V.

(b) To achieve simultaneous elimination of the third harmonic and a modulation index of $M_i = 0.8$ requires the solution to the equations

$$\cos(3\alpha_1) + \cos(3\alpha_2) = 0$$

and

$$\cos(\alpha_1) + \cos(\alpha_2) = 2M_i = 1.6$$

Using an iterative method, $\alpha_1 = 7.6^\circ$ and $\alpha_2 = 52.4^\circ$.

The preceding concept can be extended to a multilevel converter having several dc sources. For k separate sources connected in cascade, there are $2k+1$ possible voltage levels. As more dc sources and H bridges are added, the total output voltage has more steps, producing a staircase waveform that more closely approaches a sinusoid. For a five-source system as shown in Fig. 8-11, there are 11 possible output voltage levels, as illustrated in Fig. 8-12.

The Fourier series for a staircase waveform such as that in Fig 8-12 for k separate dc sources each equal to V_{dc} is

$$v_o(t) = \frac{4V_{dc}}{\pi} \sum_{n=1,3,5,7,\dots}^{\infty} [\cos(n\alpha_1) + \cos(n\alpha_2) + \dots + \cos(n\alpha_k)] \frac{\sin(n\omega_0 t)}{n} \quad (8-28)$$

The magnitudes of the Fourier coefficients are thus

$$V_n = \frac{4V_{dc}}{n\pi} [\cos(n\alpha_1) + \cos(n\alpha_2) + \dots + \cos(n\alpha_k)] \quad (8-29)$$

for $n = 1, 3, 5, 7, \dots$

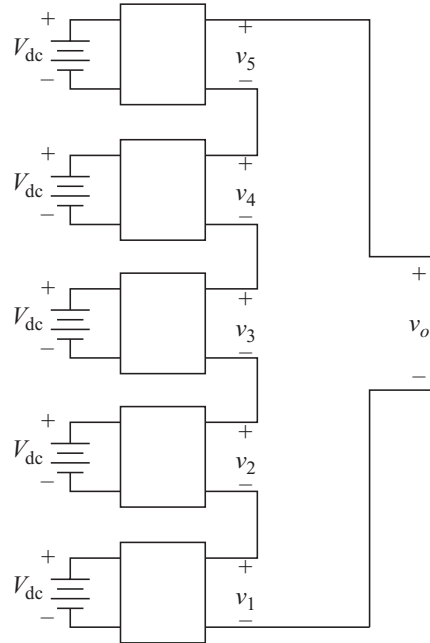


Figure 8-11 A five-source cascade multilevel converter.

The modulation index M_i for k dc sources each equal to V_{dc} is

$$M_i = \frac{V_1}{4kV_{dc}/\pi} = \frac{\cos(\alpha_1) + \cos(\alpha_2) + \cdots + \cos(\alpha_k)}{k} \quad (8-30)$$

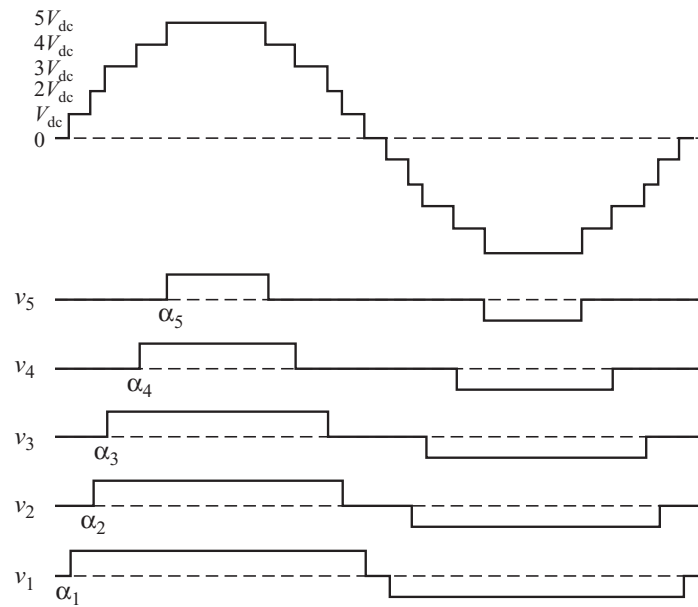


Figure 8-12 Voltages at each H bridge in Fig. 8-11 and the total output voltage.

Specific harmonics can be eliminated from the output voltage. To eliminate the m th harmonic, the delay angles must satisfy the equation

$$\cos(m\alpha_1) + \cos(m\alpha_2) + \cdots + \cos(m\alpha_k) = 0 \quad (8-31)$$

For k dc sources, $k - 1$ harmonics can be eliminated while establishing a particular M_i .

EXAMPLE 8-7

A Five-Source Multilevel Inverter

Determine the delay angles required for a five-source cascade multilevel converter that will eliminate harmonics 5, 7, 11, and 13 and will have a modulation index $M_i = 0.8$.

■ Solution

The delay angles must satisfy these simultaneous equations:

$$\begin{aligned} \cos(5\alpha_1) + \cos(5\alpha_2) + \cos(5\alpha_3) + \cos(5\alpha_4) + \cos(5\alpha_5) &= 0 \\ \cos(7\alpha_1) + \cos(7\alpha_2) + \cos(7\alpha_3) + \cos(7\alpha_4) + \cos(7\alpha_5) &= 0 \\ \cos(11\alpha_1) + \cos(11\alpha_2) + \cos(11\alpha_3) + \cos(11\alpha_4) + \cos(11\alpha_5) &= 0 \\ \cos(13\alpha_1) + \cos(13\alpha_2) + \cos(13\alpha_3) + \cos(13\alpha_4) + \cos(13\alpha_5) &= 0 \\ \cos(\alpha_1) + \cos(\alpha_2) + \cos(\alpha_3) + \cos(\alpha_4) + \cos(\alpha_5) &= 5M_i = 5(0.8) = 4 \end{aligned}$$

An iteration method such as the Newton-Raphson method must be used to solve these equations. The result is $\alpha_1 = 6.57^\circ$, $\alpha_2 = 18.94^\circ$, $\alpha_3 = 27.18^\circ$, $\alpha_4 = 45.14^\circ$, and $\alpha_5 = 62.24^\circ$. See the references in the Bibliography for information on the technique.

Equalizing Average Source Power with Pattern Swapping

In the two-source inverter of Fig. 8-9 using the switching scheme of Fig. 8-10, the source and H bridge producing the voltage v_1 supplies more average power (and energy) than the source and H bridge producing v_2 due to longer pulse widths in both the positive and negative half cycles. If the dc sources are batteries, one battery will discharge faster than the other. A technique known as pattern swapping or duty swapping equalizes the average power supplied by each dc source.

The principle of pattern swapping is to have each dc source conduct for an equal amount of time on average. An alternate switching scheme for the two-source circuit is shown in Fig. 8-13. In this scheme, the first source conducts for a longer time in the first half-cycle while the second source conducts for more time in the second half-cycle. Thus, over one complete period, the sources conduct equally, and average power from each source is the same.

For the five-source converter in Fig. 8-11, a switching scheme to equalize average power is shown in Fig. 8-14. Note that five half cycles are required to equalize power.

A variation of the H bridge multilevel inverter is to have the dc sources be of different values. The output voltage would be a staircase waveform, but not in equal

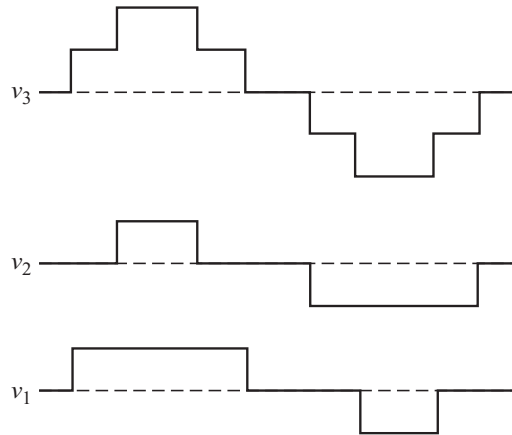


Figure 8-13 Pattern swapping to equalize average power in each source for the two-source inverter of Fig. 8-9.

voltage increments. The Fourier series of the output voltage would have different-valued harmonic amplitudes which may be an advantage in some applications.

Because independent voltage sources are needed, the multiple-source implementation of multilevel converters is best suited in applications where batteries, fuel cells, or photovoltaics are the sources.

Diode-Clamped Multilevel Inverters

A multilevel converter circuit that has the advantage of using a single dc source rather than multiple sources is the diode-clamped multilevel converter shown in Fig. 8-15a. In this circuit, the dc voltage source is connected to a pair of series capacitors, each charged to $V_{dc}/2$. The following analysis shows how the output voltage can have the levels of V_{dc} , $V_{dc}/2$, 0 , $-V_{dc}/2$, or $-V_{dc}$.

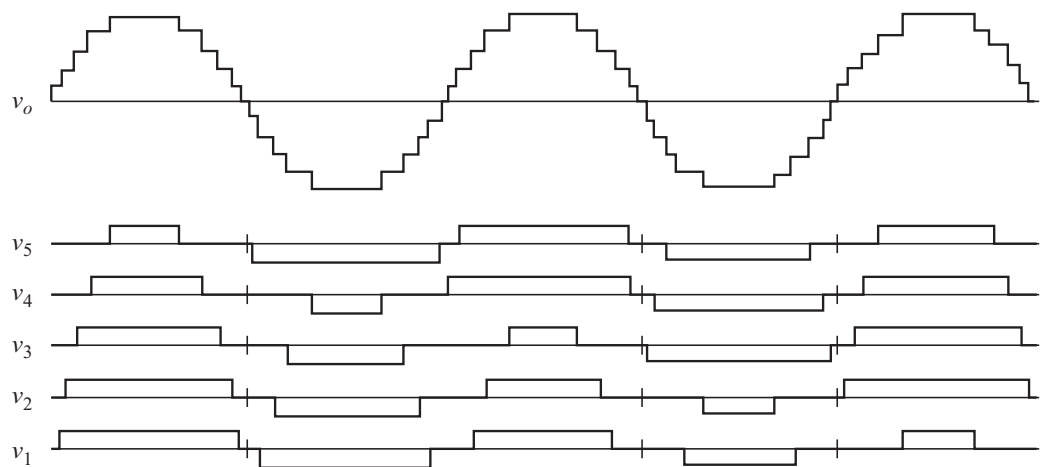


Figure 8-14 Pattern swapping to equalize average source power for the five-source multilevel inverter of Fig. 8-11.

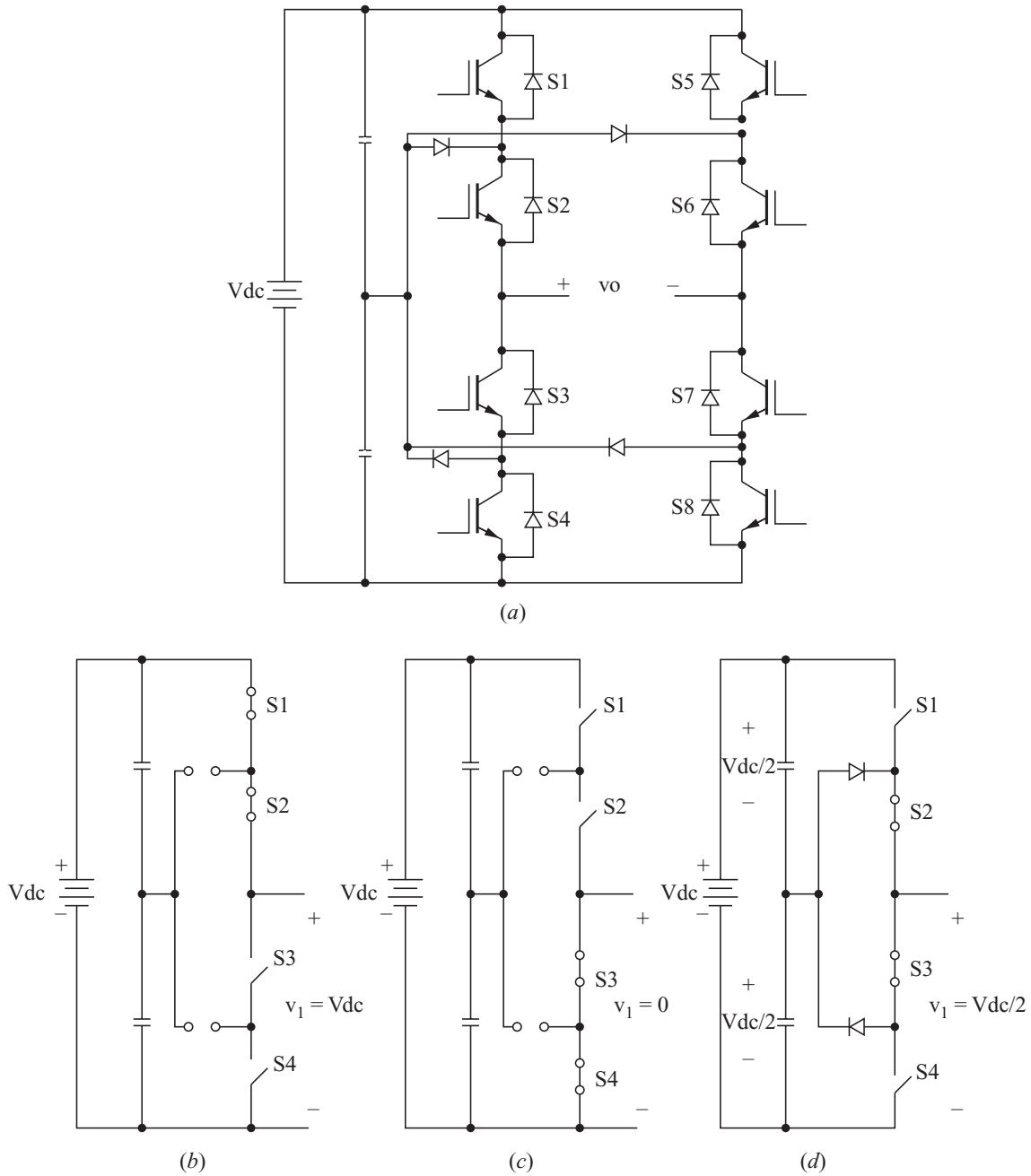


Figure 8-15 (a) A diode-clamped multilevel inverter implemented with IGBTs. (b) Analysis for one-half of the circuit for $v_1 = V_{dc}$, (c) for $v_1 = 0$, and (d) for $v_1 = \frac{1}{2}V_{dc}$.

For the analysis, consider only the left half of the bridge, as shown in Fig. 8-15b, c, and d. With S_1 and S_2 closed and S_3 and S_4 open, $V_1 = V_{dc}$ (Fig. 8-15b). The diodes are off for this condition. With S_1 and S_2 open and S_3 and S_4 closed, $V_1 = 0$ (Fig. 8-15c). The diodes are off for this condition also. To produce a voltage of $V_{dc}/2$, S_2 and S_3 are closed, and S_1 and S_4 are open (Fig. 8-15d). The voltage v_1 is that of the lower capacitor, at voltage $V_{dc}/2$, connected through the antiparallel diode path that can carry load current in either direction. Note that for each of

these circuits, two switches are open, and the voltage of the source divides between the two, thus reducing the voltage stress across each switch compared to the H bridge circuit of Fig. 8-1.

Using a similar analysis, the right half of the bridge can also produce the voltages V_{dc} , 0, and $V_{dc}/2$. The output voltage is the difference of the voltages between each half bridge, resulting in the five levels

$$v_o \in \left\{ V_{dc}, \frac{1}{2}V_{dc}, 0, -\frac{1}{2}V_{dc}, -V_{dc} \right\} \quad (8-32)$$

with multiple ways to achieve some of them. The switch control can establish delay angles α_1 and α_2 , to produce an output voltage like that in Fig. 8-10 for the cascaded H bridge, except that the maximum value is V_{dc} instead of $2V_{dc}$.

More output voltage levels are achieved with additional capacitors and switches. Figure 8-16 shows the dc source divided across three series capacitors.

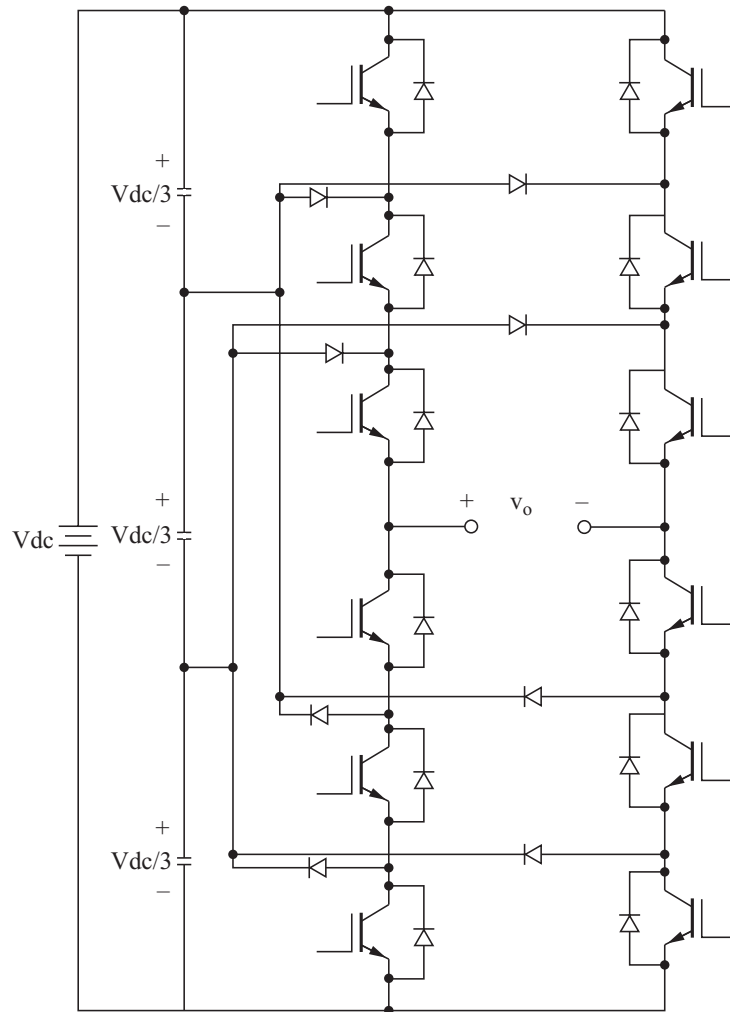


Figure 8-16 A diode-clamped multilevel inverter that produces four voltage levels on each side of the bridge and seven output voltage levels.

The voltage across each capacitor is $\frac{1}{3} V$, producing the four voltage levels on each side of the bridge of V_{dc} , $\frac{2}{3}V_{dc}$, $\frac{1}{3}V_{dc}$, and 0. The output voltage can then have the seven levels

$$v_o \in \left\{ V_{dc}, \frac{2}{3}V_{dc}, \frac{1}{3}V_{dc}, 0, -\frac{1}{3}V_{dc}, -\frac{2}{3}V_{dc}, -V_{dc} \right\} \quad (8-33)$$

8.10 PULSE-WIDTH-MODULATED OUTPUT

Pulse-width modulation (PWM) provides a way to decrease the total harmonic distortion of load current. A PWM inverter output, with some filtering, can generally meet THD requirements more easily than the square wave switching scheme. The unfiltered PWM output will have a relatively high THD, but the harmonics will be at much higher frequencies than for a square wave, making filtering easier.

In PWM, the amplitude of the output voltage can be controlled with the modulating waveforms. *Reduced filter requirements to decrease harmonics and the control of the output voltage amplitude are two distinct advantages of PWM.* Disadvantages include more complex control circuits for the switches and increased losses due to more frequent switching.

Control of the switches for sinusoidal PWM output requires (1) a reference signal, sometimes called a modulating or control signal, which is a sinusoid in this case and (2) a carrier signal, which is a triangular wave that controls the switching frequency. Bipolar and unipolar switching schemes are discussed next.

Bipolar Switching

Figure 8-17 illustrates the principle of sinusoidal bipolar pulse-width modulation. Figure 8-17a shows a sinusoidal reference signal and a triangular carrier signal. When the instantaneous value of the sine reference is larger than the triangular carrier, the output is at $+V_{dc}$, and when the reference is less than the carrier, the output is at $-V_{dc}$:

$$\begin{aligned} v_o &= +V_{dc} & \text{for } v_{\text{sine}} > v_{\text{tri}} \\ v_o &= -V_{dc} & \text{for } v_{\text{sine}} < v_{\text{tri}} \end{aligned} \quad (8-34)$$

This version of PWM is *bipolar* because the output alternates between plus and minus the dc supply voltage.

The switching scheme that will implement bipolar switching using the full-bridge inverter of Fig. 8-1 is determined by comparing the instantaneous reference and carrier signals:

$$\begin{aligned} S_1 \text{ and } S_2 \text{ are on when } v_{\text{sine}} > v_{\text{tri}} & \quad (v_o = +V_{dc}) \\ S_3 \text{ and } S_4 \text{ are on when } v_{\text{sine}} < v_{\text{tri}} & \quad (v_o = -V_{dc}) \end{aligned}$$

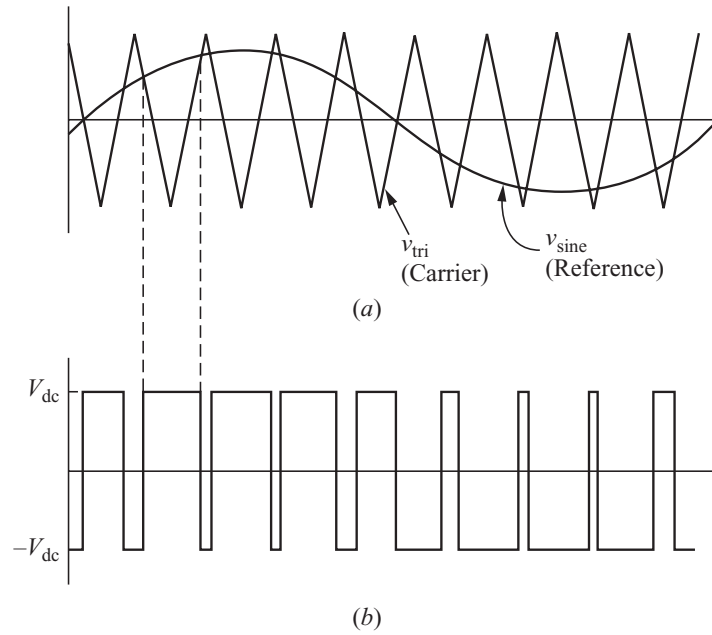


Figure 8-17 Bipolar pulse-width modulation. (a) Sinusoidal reference and triangular carrier; (b) Output is $+V_{\text{dc}}$ when $v_{\text{sine}} > v_{\text{tri}}$ and is $-V_{\text{dc}}$ when $v_{\text{sine}} < v_{\text{tri}}$.

Unipolar Switching

In a unipolar switching scheme for pulse-width modulation, the output is switched either from high to zero or from low to zero, rather than between high and low as in bipolar switching. One unipolar switching scheme has switch controls in Fig. 8-1 as follows:

$$\begin{aligned} S_1 &\text{ is on when } v_{\text{sine}} > v_{\text{tri}} \\ S_2 &\text{ is on when } -v_{\text{sine}} < v_{\text{tri}} \\ S_3 &\text{ is on when } -v_{\text{sine}} > v_{\text{tri}} \\ S_4 &\text{ is on when } v_{\text{sine}} < v_{\text{tri}} \end{aligned}$$

Note that switch pairs (S_1, S_4) and (S_2, S_3) are complementary—when one switch in a pair is closed, the other is open. The voltages v_a and v_b in Fig. 8-18a alternate between $+V_{\text{dc}}$ and zero. The output voltage $v_o = v_{ab} = v_a - v_b$ is as shown in Fig. 8-18d.

Another unipolar switching scheme has only one pair of switches operating at the carrier frequency while the other pair operates at the reference frequency, thus having two high-frequency switches and two low-frequency switches. In this switching scheme,

$$\begin{aligned} S_1 &\text{ is on when } v_{\text{sine}} > v_{\text{tri}} && \text{(high frequency)} \\ S_4 &\text{ is on when } v_{\text{sine}} < v_{\text{tri}} && \text{(high frequency)} \end{aligned}$$

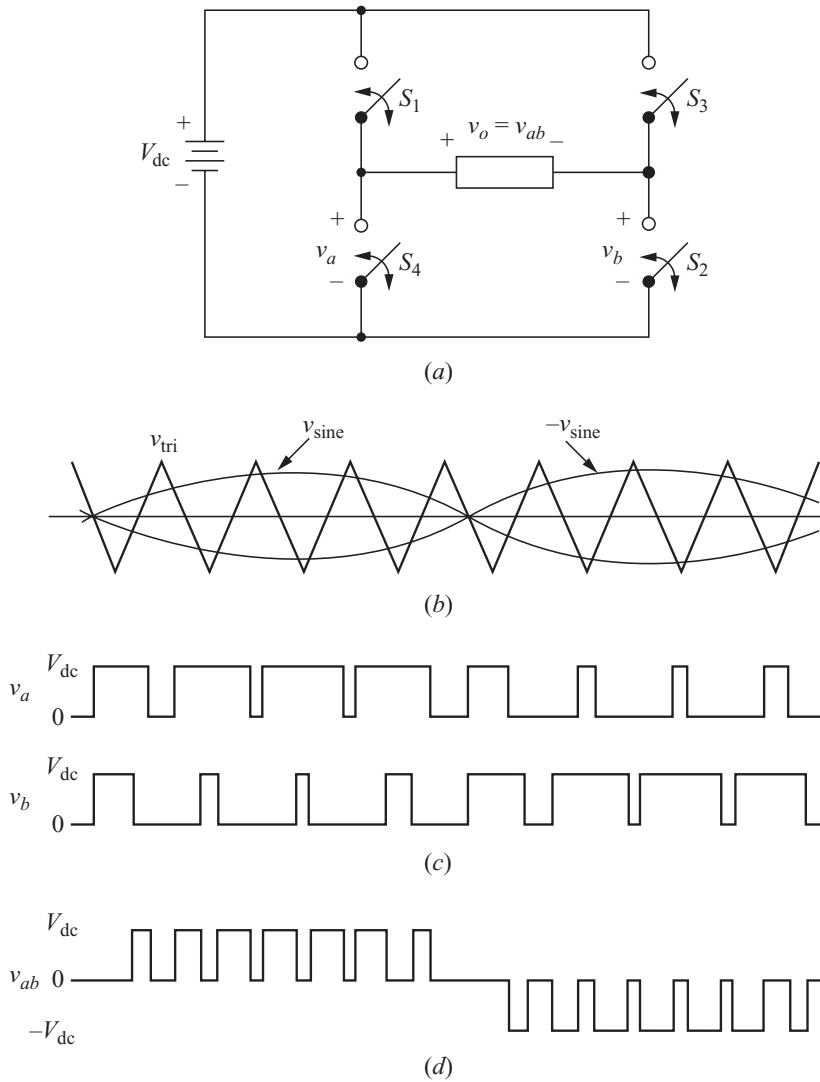


Figure 8-18 (a) Full-bridge converter for unipolar PWM; (b) Reference and carrier signals; (c) Bridge voltages v_a and v_b ; (d) Output voltage.

S_2 is on when $v_{\text{sine}} > 0$ (low frequency)

S_3 is on when $v_{\text{sine}} < 0$ (low frequency)

where the sine and triangular waves are as shown in Fig. 8-19a. Alternatively, S_2 and S_3 could be the high-frequency switches, and S_1 and S_4 could be the low-frequency switches.

8.11 PWM DEFINITIONS AND CONSIDERATIONS

At this point, some definitions and considerations relevant when using PWM should be stated.

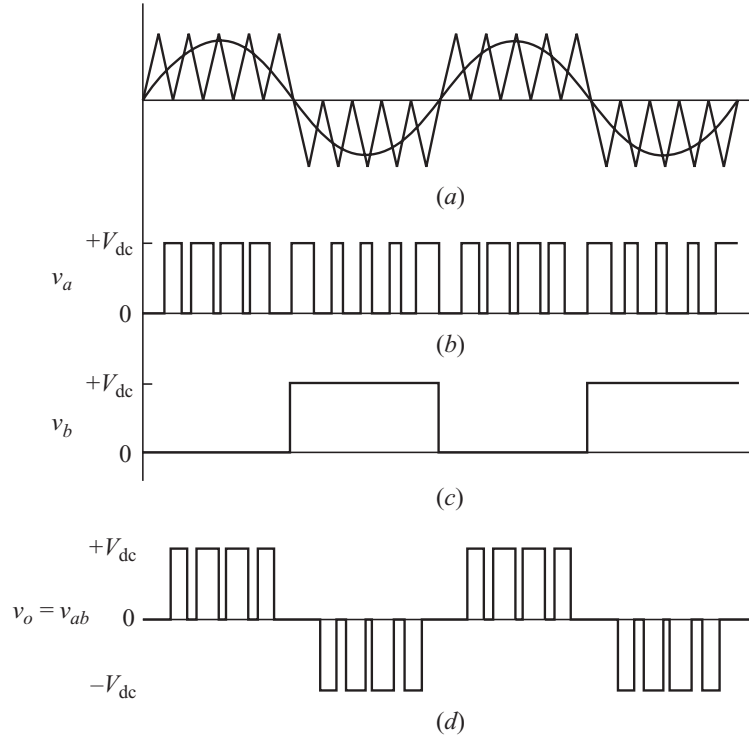


Figure 8-19 Unipolar PWM with high- and low-frequency switches. (a) Reference and control signals; (b) v_a (Fig. 8-18a); (c) v_b ; (d) output $v_a - v_b$.

1. *Frequency modulation ratio m_f* . The Fourier series of the PWM output voltage has a fundamental frequency which is the same as the reference signal. Harmonic frequencies exist at and around multiples of the switching frequency. The magnitudes of some harmonics are quite large, sometimes larger than the fundamental. However, because these harmonics are located at high frequencies, a simple low-pass filter can be quite effective in removing them. Details of the harmonics in PWM are given in the next section. The *frequency modulation ratio m_f* is defined as the ratio of the frequencies of the carrier and reference signals,

$$m_f = \frac{f_{\text{carrier}}}{f_{\text{reference}}} = \frac{f_{\text{tri}}}{f_{\text{sine}}} \quad (8-35)$$

Increasing the carrier frequency (increasing m_f) increases the frequencies at which the harmonics occur. A disadvantage of high switching frequencies is higher losses in the switches used to implement the inverter.

2. *Amplitude modulation ratio m_a* . The *amplitude modulation ratio m_a* is defined as the ratio of the amplitudes of the reference and carrier signals:

$$m_a = \frac{V_{m, \text{reference}}}{V_{m, \text{carrier}}} = \frac{V_{m, \text{sine}}}{V_{m, \text{tri}}} \quad (8-36)$$

If $m_a \leq 1$, the amplitude of the fundamental frequency of the output voltage V_1 is linearly proportional to m_a . That is,

$$V_1 = m_a V_{dc} \quad (8-37)$$

The amplitude of the fundamental frequency of the PWM output is thus controlled by m_a . This is significant in the case of an unregulated dc supply voltage because the value of m_a can be adjusted to compensate for variations in the dc supply voltage, producing a constant-amplitude output. Alternatively, m_a can be varied to change the amplitude of the output. If m_a is greater than 1, the amplitude of the output increases with m_a , but not linearly.

3. *Switches.* The switches in the full-bridge circuit must be capable of carrying current in either direction for pulse-width modulation just as they did for square wave operation. Feedback diodes across the switching devices are necessary, as was done in the inverter in Fig. 8-3a. Another consequence of real switches is that they do not turn on or off instantly. Therefore, it is necessary to allow for switching times in the control of the switches just as it was for the square-wave inverter.
4. *Reference voltage.* The sinusoidal reference voltage must be generated within the control circuit of the inverter or taken from an outside reference. It may seem as though the function of the inverter bridge is unnecessary because a sinusoidal voltage must be present before the bridge can operate to produce a sinusoidal output. However, there is very little power required from the reference signal. The power supplied to the load is provided by the dc power source, and this is the intended purpose of the inverter. The reference signal is not restricted to a sinusoid, and other waveshapes can function as the reference signal.

8.12 PWM HARMONICS

Bipolar Switching

The Fourier series of the bipolar PWM output illustrated in Fig. 8-17 is determined by examining each pulse. The triangular waveform is synchronized to the reference as shown in Fig 8-17a, and m_f is chosen to be an odd integer. The PWM output then exhibits odd symmetry, and the Fourier series can then be expressed

$$v_o(t) = \sum_{n=1}^{\infty} V_n \sin(n\omega_0 t) \quad (8-38)$$

For the k th pulse of the PWM output in Fig. 8-20, the Fourier coefficient is

$$\begin{aligned} V_{nk} &= \frac{2}{\pi} \int_0^T v(t) \sin(n\omega_0 t) d(\omega_0 t) \\ &= \frac{2}{\pi} \left[\int_{\alpha_k}^{\alpha_k + \delta_k} V_{dc} \sin(n\omega_0 t) d(\omega_0 t) + \int_{\alpha_k + \delta_k}^{\alpha_{k+1}} -(V_{dc}) \sin(n\omega_0 t) (d(\omega_0 t)) \right] \end{aligned}$$

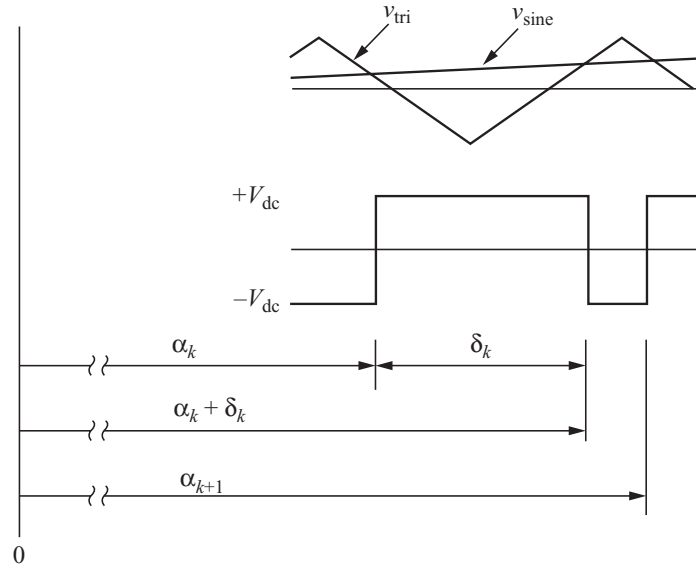


Figure 8-20 Single PWM pulse for determining Fourier series for bipolar PWM.

Performing the integration,

$$V_{nk} = \frac{2V_{dc}}{n\pi} [\cos n\alpha_k + \cos n\alpha_{k+1} - 2\cos n(\alpha_k + \delta_k)] \quad (8-39)$$

Each Fourier coefficient V_n for the PWM waveform is the sum of V_{nk} for the p pulses over one period,

$$V_n = \sum_{k=1}^p V_{nk} \quad (8-40)$$

The normalized frequency spectrum for bipolar switching for $m_a=1$ is shown in Fig. 8-21. The harmonic amplitudes are a function of m_a because the

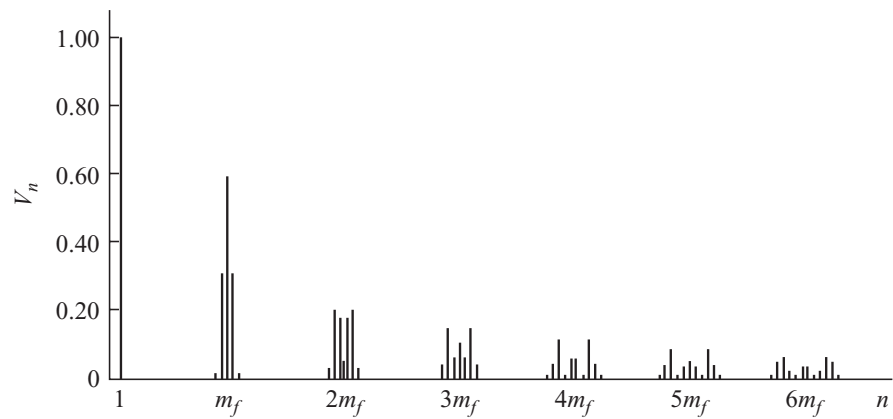


Figure 8-21 Frequency spectrum for bipolar PWM with $m_a = 1$.

Table 8-3 Normalized Fourier Coefficients V_n/V_{dc} for Bipolar PWM

	$m_a=1$	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
$n=1$	1.00	0.90	0.80	0.70	0.60	0.50	0.40	0.30	0.20	0.10
$n=m_f$	0.60	0.71	0.82	0.92	1.01	1.08	1.15	1.20	1.24	1.27
$n=m_f \pm 2$	0.32	0.27	0.22	0.17	0.13	0.09	0.06	0.03	0.02	0.00

width of each pulse depends on the relative amplitudes of the sine and triangular waves. The first harmonic frequencies in the output spectrum are at and around m_f . Table 8-3 indicates the first harmonics in the output for bipolar PWM. The Fourier coefficients are not a function of m_f if m_f is large (≥ 9).

EXAMPLE 8-8**A PWM Inverter**

The full-bridge inverter is used to produce a 60-Hz voltage across a series RL load using bipolar PWM. The dc input to the bridge is 100 V, the amplitude modulation ratio m_a is 0.8, and the frequency modulation ratio m_f is 21 [$f_{tri} = (21)(60) = 1260$ Hz]. The load has a resistance of $R = 10 \Omega$ and series inductance $L = 20$ mH. Determine (a) the amplitude of the 60-Hz component of the output voltage and load current, (b) the power absorbed by the load resistor, and (c) the THD of the load current.

■ Solution

(a) Using Eq. (8-38) and Table 8-3, the amplitude of the 60-Hz fundamental frequency is

$$V_1 = m_a V_{dc} = (0.8)(100) = 80 \text{ V}$$

The current amplitudes are determined using phasor analysis:

$$I_n = \frac{V_n}{Z_n} = \frac{V_n}{\sqrt{R^2 + (n\omega_0 L)^2}} \quad (8-41)$$

For the fundamental frequency,

$$I_1 = \frac{80}{\sqrt{10^2 + [(1)(2\pi 60)(0.02)]^2}} = 6.39 \text{ A}$$

(b) With $m_f = 21$, the first harmonics are at $n = 21, 19$, and 23 . Using Table 8-3,

$$V_{21} = (0.82)(100) = 82 \text{ V}$$

$$V_{19} = V_{23} = (0.22)(100) = 22 \text{ V}$$

Current at each of the harmonics is determined from Eq. (8-41).

Power at each frequency is determined from

$$P_n = (I_{n, \text{rms}})^2 R = \left(\frac{I_n}{\sqrt{2}} \right)^2 R$$

Table 8-4 Fourier Series Quantities for the PWM Inverter of Example 8-8

n	f_n (Hz)	V_n (V)	Z_n (Ω)	I_n (A)	$I_{n,\text{rms}}$ (A)	P_n (W)
1	60	80.0	12.5	6.39	4.52	204.0
19	1140	22.0	143.6	0.15	0.11	0.1
21	1260	81.8	158.7	0.52	0.36	1.3
23	1380	22.0	173.7	0.13	0.09	0.1

The resulting voltage amplitudes, currents, and powers at these frequencies are summarized in Table 8-4.

Power absorbed by the load resistor is

$$P = \sum P_n \approx 204.0 + 0.1 + 1.3 + 0.1 = 205.5 \text{ W}$$

Higher-order harmonics contribute little power and can be neglected.

- (c) The THD of the load current is determined using Eq. (8-17) with the rms current of the harmonics approximated by the first few terms indicated in Table 8-4.

$$\text{THD}_1 = \frac{\sqrt{\sum_{n=2}^{\infty} I_{n,\text{rms}}^2}}{I_{1,\text{rms}}} \approx \frac{\sqrt{(0.11)^2 + (0.36)^2 + (0.09)^2}}{4.52} = 0.087 = 8.7\%$$

By using the truncated Fourier series in Table 8-4, the THD will be underestimated. However, since the impedance of the load increases and the amplitudes of the harmonics generally decrease as n increases, the above approximation should be acceptable. (Including currents through $n = 100$ gives a THD of 9.1 percent.)

EXAMPLE 8-9

PWM Inverter Design

Design a bipolar PWM inverter that will produce a 75-V rms 60-Hz output from a 150-V dc source. The load is a series RL combination with $R = 12 \Omega$ and $L = 60 \text{ mH}$. Select the switching frequency such that the current THD is less than 10 percent.

■ Solution

The required amplitude modulation ratio is determined from Eq. (8-38),

$$m_a = \frac{V_1}{V_{dc}} = \frac{75\sqrt{2}}{150} = 0.707$$

The current amplitude at 60 Hz is

$$I_1 = \frac{V_1}{Z_1} = \frac{75\sqrt{2}}{\sqrt{12^2 + [(2\pi 60)(0.06)]^2}} = 4.14 \text{ A}$$

The rms value of the harmonic current has a limit imposed by the required THD,

$$\sqrt{\sum_{n=2}^{\infty} (I_{n,\text{rms}})^2} < 0.1 I_{1,\text{rms}} = 0.1 \left(\frac{4.14}{\sqrt{2}} \right) = 0.293 \text{ A}$$

The term that will produce the dominant harmonic current is at the switching frequency. As an approximation, assume that the harmonic content of the load current is the same as the dominant harmonic at the carrier frequency:

$$\sqrt{\sum_{n=2}^{\infty} (I_{n,\text{rms}})^2} \approx I_{mf,\text{rms}} = \frac{I_{mf}}{\sqrt{2}}$$

The amplitude of the current harmonic at the carrier frequency is then approximated as

$$I_{mf} < (0.1)(4.14) = 0.414 \text{ A}$$

Table 8-3 indicates that the normalized voltage harmonic for $n = m_f$ and for $m_a = 0.7$ is 0.92. The voltage amplitude for $n = m_f$ is then

$$V_{mf} = 0.92 V_{dc} = (0.92)(150) = 138 \text{ V}$$

The minimum load impedance at the carrier frequency is then

$$Z_{mf} = \frac{V_{mf}}{I_{mf}} = \frac{138}{0.414} = 333 \text{ } \Omega$$

Because the impedance at the carrier frequency must be much larger than the 12- Ω load resistance, assume the impedance at the carrier frequency is entirely inductive reactance,

$$Z_{mf} \approx \omega L = m_f \omega_0 L$$

For the load impedance to be greater than 333 Ω ,

$$\begin{aligned} m_f \omega_0 L &> 333 \\ m_f &> \frac{333}{(377)(0.06)} = 14.7 \end{aligned}$$

Selecting m_f to be at least 15 would marginally meet the design specifications. However, the estimate of the harmonic content used in the calculations will be low, so a higher carrier frequency is a more prudent selection. Let $m_f = 17$, which is the next odd integer. The carrier frequency is then

$$f_{\text{tri}} = m_f f_{\text{ref}} = (17)(60) = 1020 \text{ Hz}$$

Further increasing m_f would reduce the current THD, but at the expense of larger switching losses. A PSpice simulation, as discussed later in this chapter, can be used to verify that the design meets the specifications.

Unipolar Switching

With the unipolar switching scheme in Fig. 8-18, some harmonics that were in the spectrum for the bipolar scheme are absent. The harmonics in the output begin at around $2m_f$, and m_f is chosen to be an even integer. Figure 8-22 shows the frequency spectrum for unipolar switching with $m_a = 1$.

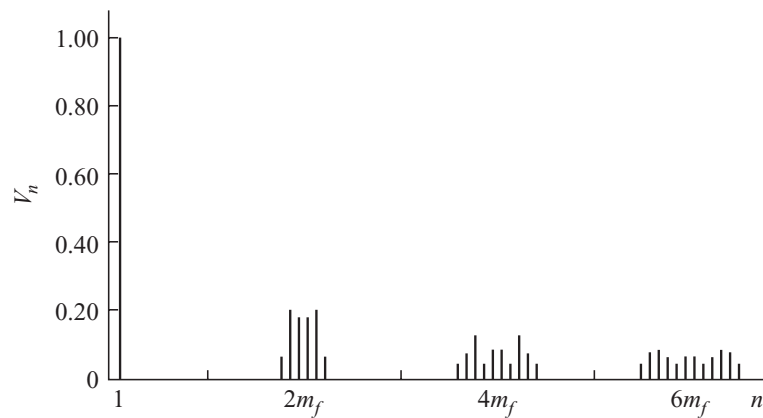


Figure 8-22 Frequency spectrum for unipolar PWM with $m_a = 1$.

Table 8-5 indicates the first harmonics in the output for unipolar PWM.

The unipolar PWM scheme using high- and low-frequency switches shown in Fig. 8-19 will have similar results as indicated above, but the harmonics will begin at around m_f rather than $2m_f$.

Table 8-5 Normalized Fourier Coefficients V_n/V_{dc} for Unipolar PWM in Fig. 8-18

	$m_a=1$	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
$n=1$	1.00	0.90	0.80	0.70	0.60	0.50	0.40	0.30	0.20	0.10
$n=2m_f\pm 1$	0.18	0.25	0.31	0.35	0.37	0.36	0.33	0.27	0.19	0.10
$n=2m_f\pm 3$	0.21	0.18	0.14	0.10	0.07	0.04	0.02	0.01	0.00	0.00

8.13 CLASS D AUDIO AMPLIFIERS

The reference signal for the PWM control circuit can be an audio signal, and the full-bridge circuit could be used as a PWM audio amplifier. A PWM audio amplifier is referred to as a *class D amplifier*. The triangular wave carrier signal for this application is typically 250 kHz to provide adequate sampling, and the PWM waveform is low-pass filtered to recover the audio signal and deliver power to a speaker. The spectrum of the PWM output signal is dynamic in this case.

Class D amplifiers are much more efficient than other types of audio power amplifiers. The class AB amplifier, the traditional circuit for audio applications, has a maximum theoretical efficiency of 78.5 percent for a sine wave of maximum undistorted output. In practice, with real audio signals, class AB efficiency is much lower, on the order of 20 percent. The theoretical efficiency of the class D amplifier is 100 percent because the transistors are used as switches. Because transistor switching and filtering are imperfect, practical class D amplifiers are about 75 percent efficient.

Class D audio amplifiers are becoming more prevalent in consumer electronics applications where greater efficiency results in reduced size and increased

battery life. In high-power applications such as at rock concerts, class D amplifiers are used to reduce the size of the amplifier and for reduced heat requirements in the equipment.

8.14 SIMULATION OF PULSE-WIDTH-MODULATED INVERTERS

Bipolar PWM

PSpice can be used to simulate the PWM inverter switching schemes presented previously in this chapter. As with other power electronics circuits, the level of circuit detail depends on the objective of the simulation. If only the voltages and currents in the load are desired, a PWM source may be created without modeling the individual switches in the bridge circuit. Figure 8-23 shows two ways to produce a bipolar PWM voltage. The first uses an ABM2 block, and the second uses

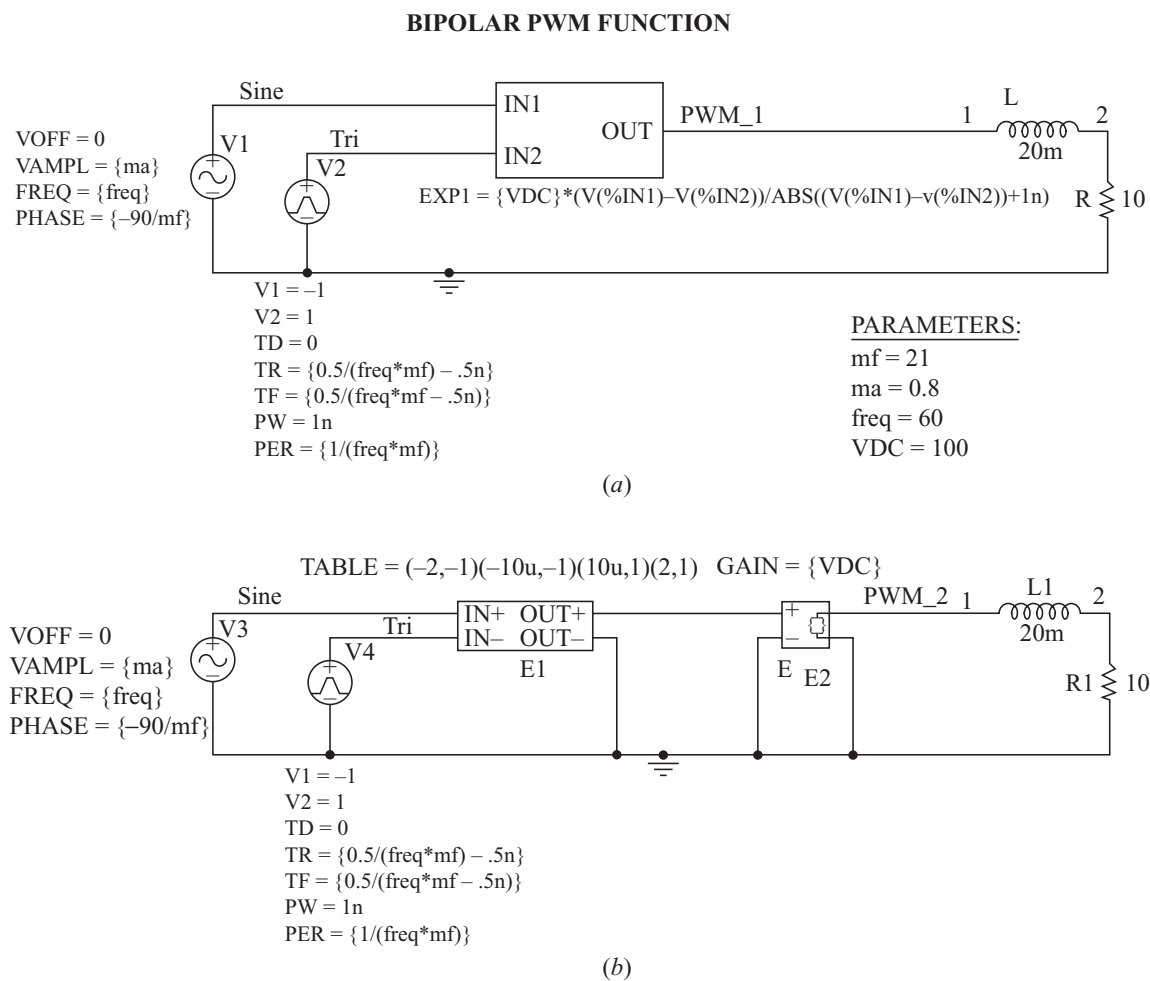


Figure 8-23 PSpice functional circuits for producing a bipolar PWM voltage using (a) an ABM block and (b) an ETABLE voltage source.

a voltage-controlled voltage source ETABLE. Both methods compare a sine wave to a triangular wave. Either method allows the behavior of a specific load to a PWM input to be investigated.

If the load contains an inductance and/or capacitance, there will be an initial transient in the load current. Since the steady-state load current is usually of interest, one or more periods of the load current must be allowed to run before meaningful output is obtained. One way to achieve this in PSpice is to delay output in the transient command, and another way is to restrict the data to steady-state results in Probe. The reference signal is synchronized with the carrier signal as in Fig. 8-17a. When the triangular carrier voltage has negative slope going through zero, the sinusoidal reference voltage must have positive slope going through zero. The triangular waveform starts at the positive peak with negative slope. The phase angle of the reference sinusoid is adjusted to make the zero crossing correspond to that of the triangular wave by using a phase angle of $-90^\circ/m_f$. The following example illustrates a PSpice simulation of a bipolar PWM application.

EXAMPLE 8-10

PSpice Simulation of PWM

Use PSpice to analyze the PWM inverter circuit of Example 8-8.

■ Solution

Using either PWM circuit in Fig. 8-23, the Probe output will be the waveforms shown in Fig. 8-24a. The current is scaled by a factor of 10 to show more clearly its relationship

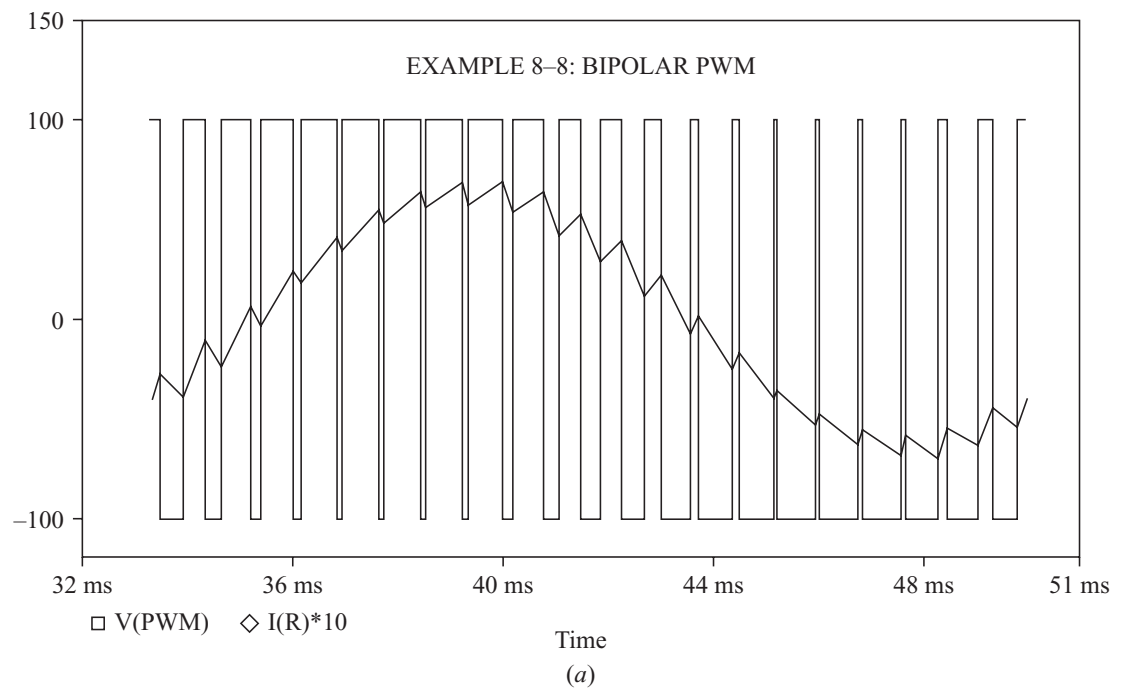


Figure 8-24 (a) Probe output for Example 8-10 showing PWM voltage and load current (current is scaled for illustration); (b) Frequency spectra for voltage and current.

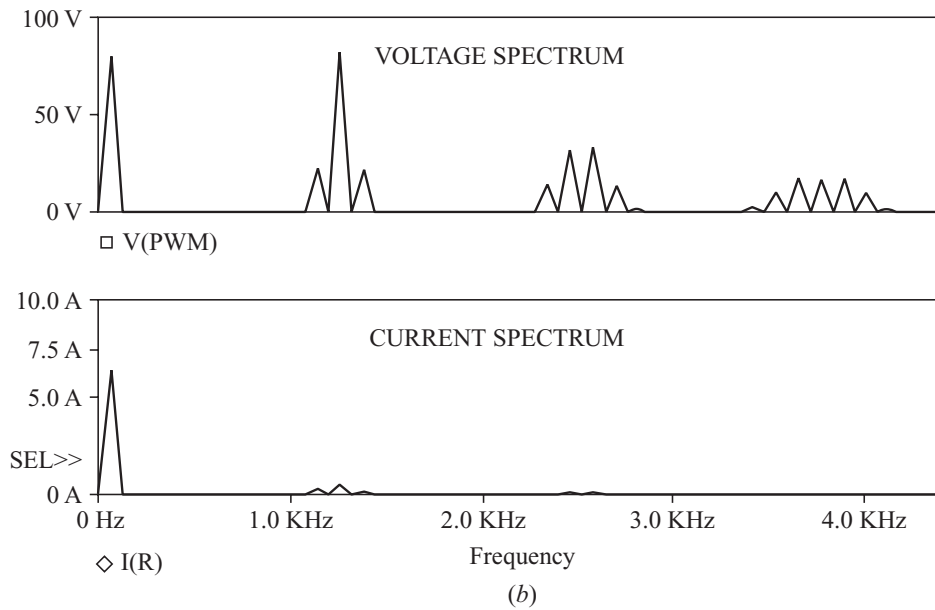


Figure 8-24 (continued)

with output voltage. Note the sinelike quality of the current. The Fourier coefficients of voltage and current are determined by using the Fourier option under the x axis menu or by pressing the FFT icon. Figure 8-24b shows the frequency spectra of voltage and current with the range on the x axis selected to show the lower frequencies. The cursor option is used to determine the Fourier coefficients.

Table 8-6 summarizes the results. Note the close correspondence with the results of Example 8-8.

Table 8-6 PSpice Results of Example 8-10

n	f_n (Hz)	V_n (V)	I_n (A)
1	60	79.8	6.37
19	1140	21.8	0.15
21	1260	82.0	0.52
23	1380	21.8	0.13

If the voltages and currents in the source and switches are desired, the PSpice input file must include the switches. A somewhat idealized circuit using voltage-controlled switches with feedback diodes is shown in Fig. 8-25. To simulate pulse-width modulation, the control for the switches in the inverter is the voltage difference between a triangular carrier voltage and a sine reference voltage. While this does not represent a model for real switches, this circuit is useful to simulate either bipolar or unipolar PWM. A more realistic bridge model would include devices such as BJTs, MOSFETs, or IGBTs for the switches. The model that is appropriate will depend on how completely switch performance must be investigated.

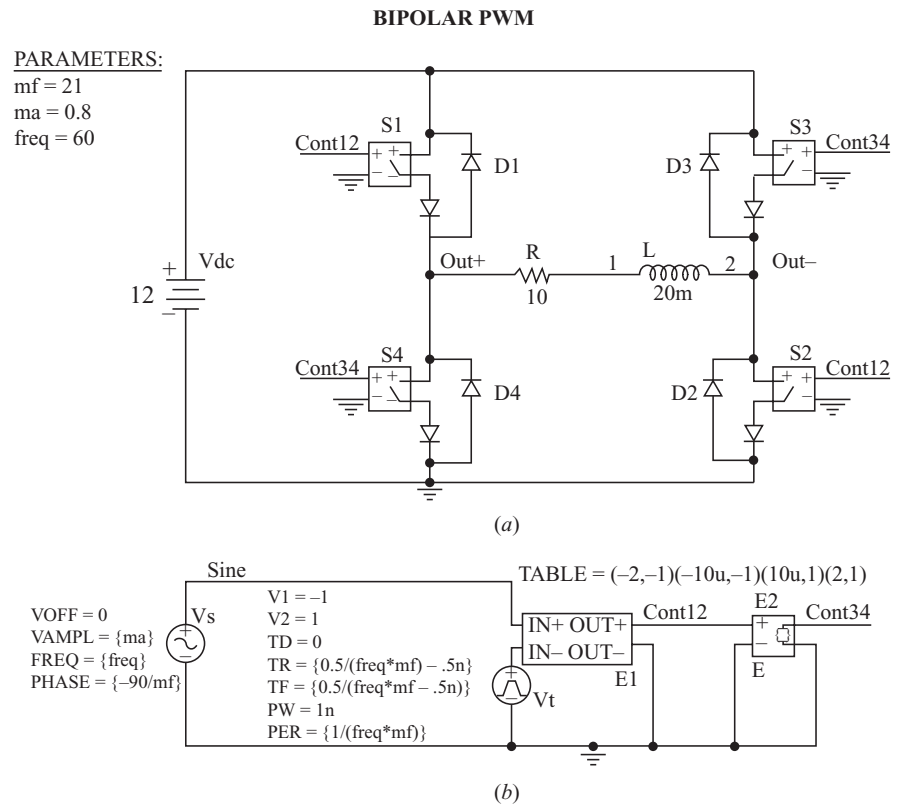


Figure 8-25 PSpice circuits for a PWM inverter (a) using voltage-controlled switches and diodes but requires the full PSpice version and (b) generating a PWM function.

Unipolar PWM

Again, unipolar PWM can be simulated using various levels of switch models. The input file shown in Fig. 8-26 utilizes dependent sources to produce a unipolar PWM output.

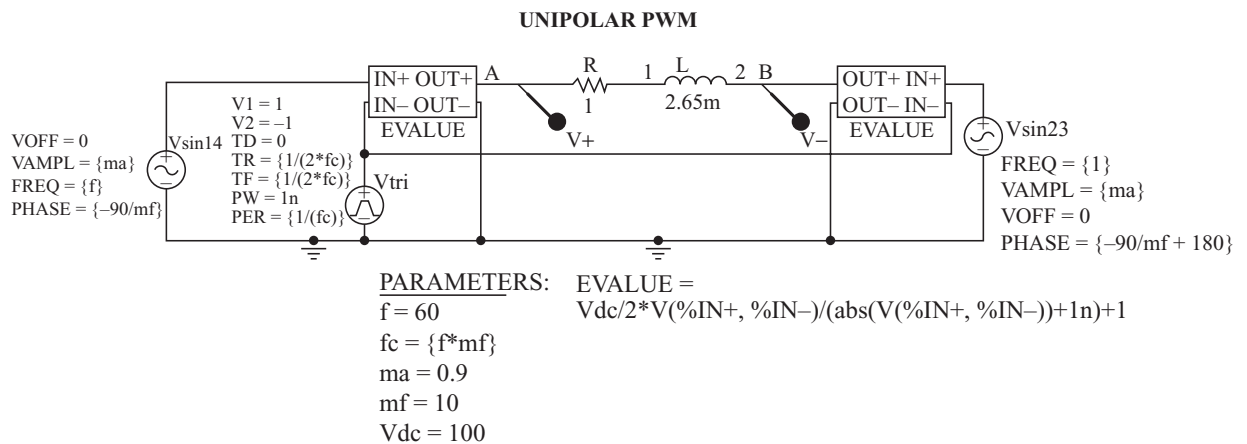


Figure 8-26 A PSpice circuit for generating a unipolar PWM voltage. The output voltage is between nodes A and B.

EXAMPLE 8-11

Pulse-Width Modulation PSpice Simulation

Pulse-width modulation is used to provide a 60-Hz voltage across a series RL load with $R = 1\ \Omega$ and $L = 2.65\text{ mH}$. The dc supply voltage is 100 V. The amplitude of the 60-Hz voltage is to be 90 V, requiring $m_a = 0.9$. Use PSpice to obtain the current waveform in the load and the THD of the current waveform in the load. Use (a) bipolar PWM with $m_f = 21$, (b) bipolar PWM with $m_f = 41$, and (c) unipolar PWM with $m_f = 10$.

■ Solution

- (a) The PSpice circuit for bipolar PWM (Fig. 8-25b) is run with $m_a = 0.9$ and $m_f = 21$. The voltage across the load and the current in the load resistor are shown in Fig. 8-27a. The currents for the 60-Hz fundamental and the lowest-order harmonics are obtained from the Fourier option under x axis in Probe. The harmonic amplitudes correspond to the peaks, and the cursor option determines precise values. The rms current can be obtained from Probe by entering the expression $\text{RMS}(I(R))$. The total harmonic distortion based on the truncated Fourier series is computed from Eq. (8-17). Results are in the table in this example.
- (b) The PSpice circuit is modified for $m_f = 41$. The voltage and current waveforms are shown in Fig. 8-27b. The resulting harmonic currents are obtained from the Fourier option in Probe.

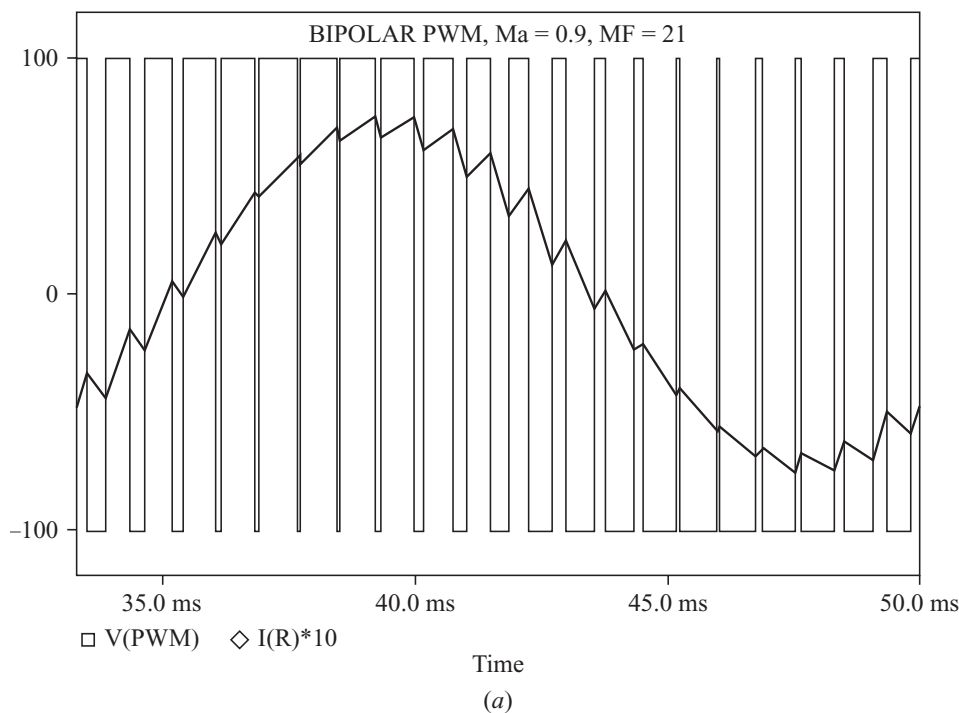


Figure 8-27 Voltage and current for Example 8-11 for (a) bipolar PWM with $m_f = 21$, (b) bipolar PWM with $m_f = 41$, (c) Unipolar PWM with $m_f = 10$.

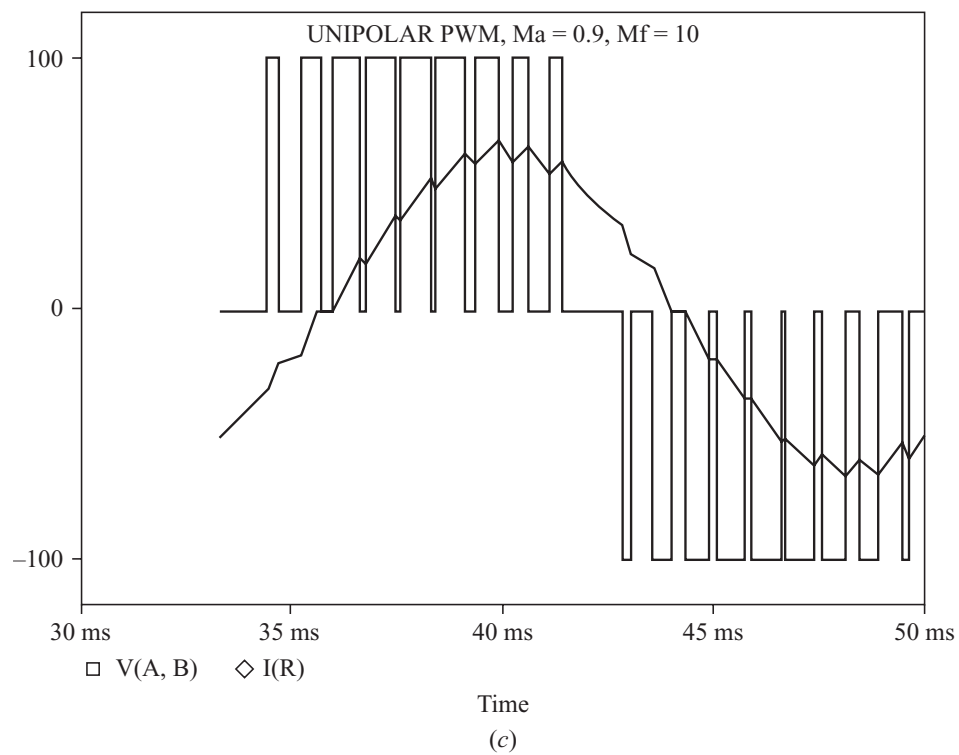
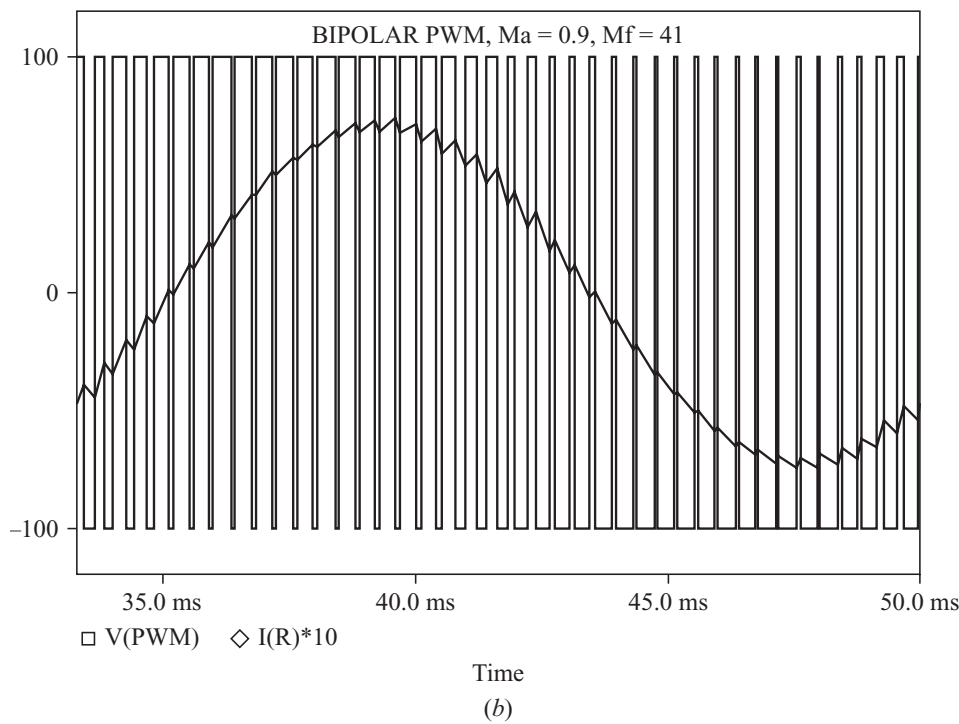


Figure 8-27 (continued)

- (c) The PSpice input file for unipolar switching in Fig. 8-26 is run with the parameter $m_f = 10$. The output voltage and current are shown in Fig. 8-27c. The results of the three simulations for this example are shown in the following table.

	Bipolar $m_f = 21$		Bipolar $m_f = 41$		Unipolar $m_f = 10$	
	f_n	I_n	f_n	I_n	f_n	I_n
	60	63.6	60	64.0	60	62.9
	1140	1.41	2340	0.69	1020	1.0
	1260	3.39	2460	1.7	1140	1.4
	1380	1.15	2580	0.62	1260	1.24
					1380	0.76
I_{rms}		45.1		45.0		44.5
THD		6.1%		3.2%		3.6%

Note that the THD is relatively low in each of these PWM switching schemes, and increasing the switching frequency (increasing m_f) decreases the harmonic currents in this type of a load.

8.15 THREE-PHASE INVERTERS

The Six-Step Inverter

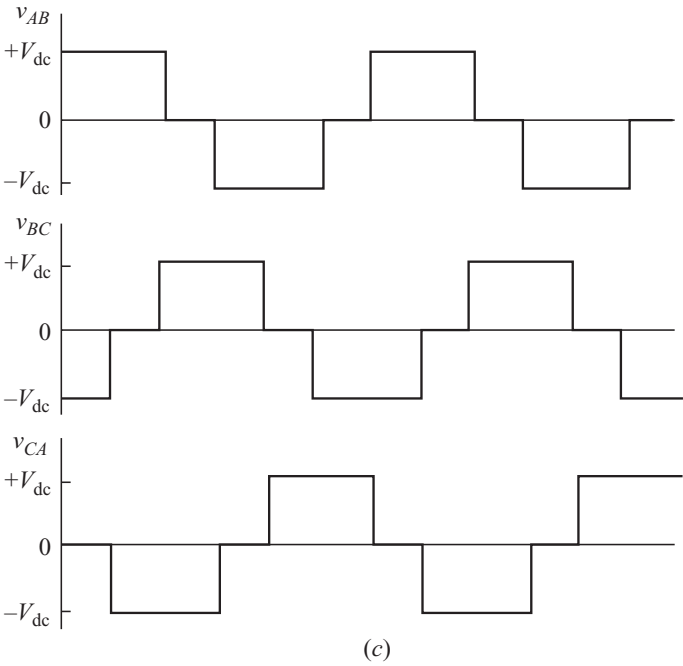
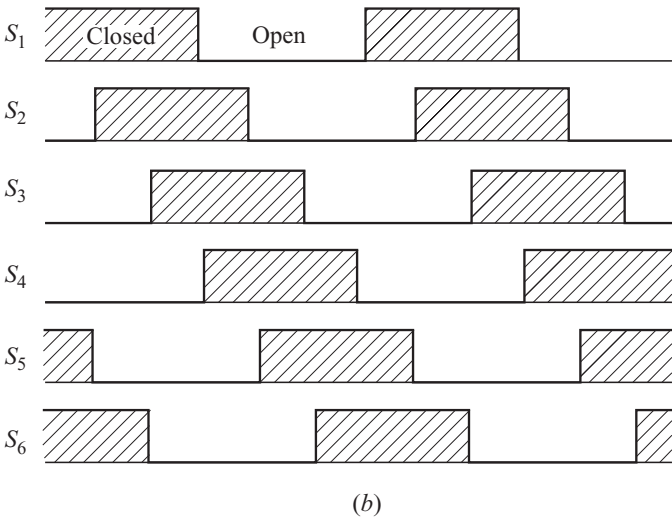
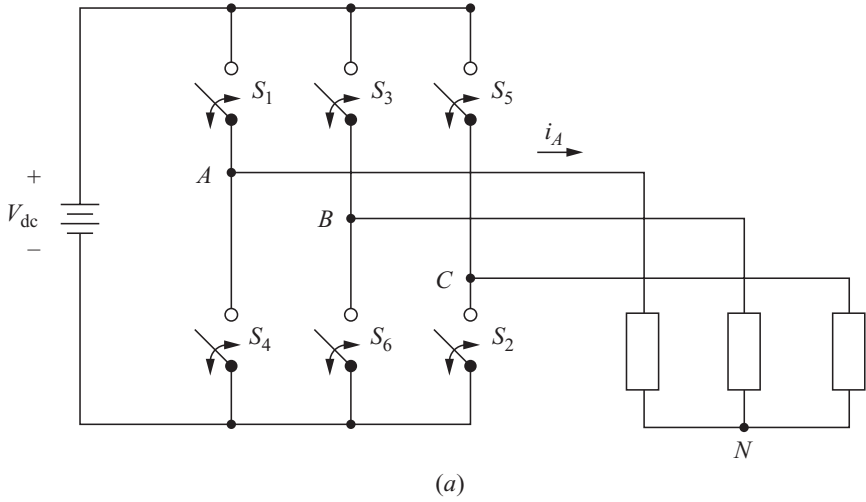
Figure 8-28a shows a circuit that produces a three-phase ac output from a dc input. A major application of this circuit is speed control of induction motors, where the output frequency is varied. The switches are closed and opened in the sequence shown in Fig. 8-28b.

Each switch has a duty ratio of 50 percent (not allowing for blanking time), and a switching action takes place every $T/6$ time interval, or 60° angle interval. Note that switches S_1 and S_4 close and open opposite of each other, as do switch pairs (S_2, S_5) and (S_3, S_6) . As with the single-phase inverter, these switch pairs must coordinate so they are not closed at the same time, which would result in a short circuit across the source. With this scheme, the instantaneous voltages v_{AO} , v_{BO} , and v_{CO} are $+V_{dc}$ or zero, and line-to-line output voltages v_{AB} , v_{BC} , and v_{CA} are $+V_{dc}$, 0, or $-V_{dc}$. The switching sequence in Fig. 8-28b produces the output voltages shown in Fig. 8-28c.

The three-phase load connected to this output voltage may be connected in delta or ungrounded neutral wye. For a wye-connected load, which is the more common load connection, the voltage across each phase of the load is a line-to-neutral voltage, shown in Fig. 8-28d. Because of the six steps in the output waveforms for the line-to-neutral voltage resulting from the six switching transitions per period, this circuit with this switching scheme is called a *six-step inverter*.

The Fourier series for the output voltage has a fundamental frequency equal to the switching frequency. Harmonic frequencies are of order $6k \pm 1$ for $k = 1, 2, \dots$ ($n = 5, 7, 11, 13 \dots$). The third harmonic and multiples of the third do not

Figure 8-28 (a) Three-phase inverter; (b) Switching sequence for six-step output; (c) Line-to-line output voltages; (d) Line-to-neutral voltages for an ungrounded Y-connected load; (e) Current in phase *A* for an *RL* load.



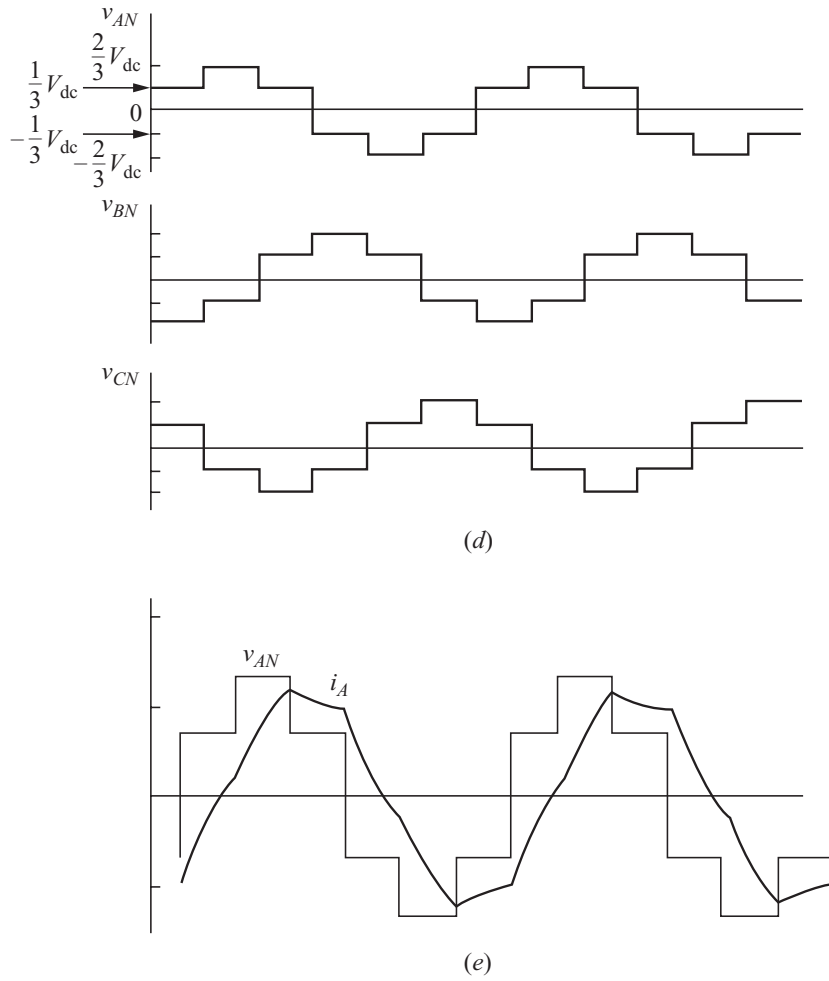


Figure 8-28 (continued)

exist, and even harmonics do not exist. For an input voltage of V_{dc} , the output for an ungrounded wye-connected load has the following Fourier coefficients:

$$V_{n,L-L} = \left| \frac{4V_{dc}}{n\pi} \cos\left(n\frac{\pi}{6}\right) \right|$$

$$V_{n,L-N} = \left| \frac{2V_{dc}}{3n\pi} \left[2 + \cos\left(n\frac{\pi}{3}\right) - \cos\left(n\frac{2\pi}{3}\right) \right] \right| \quad n = 1, 5, 7, 11, 13, \dots \quad (8-42)$$

The THD of both the line-to-line and line-to-neutral voltages can be shown to be 31 percent from Eq. (8-17). The THD of the currents in load-dependent 15 are smaller for an RL load. An example of the line-to-neutral voltage and line current for an RL wye-connected load is shown in Fig. 8-28e.

The output frequency can be controlled by changing the switching frequency. The magnitude of the output voltage depends on the value of the dc supply voltage. To control the output voltage of the six-step inverter, the dc input voltage must be adjusted.

EXAMPLE 8-12

Six-Step Three-Phase Inverter

For the six-step three-phase inverter of Fig. 8-28a, the dc input is 100 V and the fundamental output frequency is 60 Hz. The load is wye-connected with each phase of the load a series RL connection with $R = 10\ \Omega$ and $L = 20\text{ mH}$. Determine the total harmonic distortion of the load current.

■ Solution

The amplitude of load current at each frequency is

$$I_n = \frac{V_{n,L-N}}{Z_n} = \frac{V_{n,L-N}}{\sqrt{R^2 + (n\omega_0 L)^2}} = \frac{V_{n,L-N}}{\sqrt{10^2 + [n(2\pi 60)(0.02)]^2}}$$

where $V_{n,L-N}$ is determined from Eq. (8-42). Table 8-7 summarizes the results of the Fourier series computation.

Table 8.7 Fourier Components for the Six-Step Inverter of Example 8-12

n	$V_{n,L-N}$ (V)	Z_n (Ω)	I_n (A)	$I_{n,\text{rms}}$ (A)
1	63.6	12.5	5.08	3.59
5	12.73	39.0	0.33	0.23
7	9.09	53.7	0.17	0.12
11	5.79	83.5	0.07	0.05
13	4.90	98.5	0.05	0.04

The THD of the load current is computed from Eq. (8-17) as

$$\text{THD}_1 = \frac{\sqrt{\sum_{n=2}^{\infty} I_{n,\text{rms}}^2}}{I_{1,\text{rms}}} \approx \frac{\sqrt{(0.23)^2 + (0.12)^2 + (0.05)^2 + (0.04)^2}}{3.59} = 0.07 = 7\%$$

PWM Three-Phase Inverters

Pulse-width modulation can be used for three-phase inverters as well as for single-phase inverters. The advantages of PWM switching are the same as for the single-phase case: reduced filter requirements for harmonic reduction and the controllability of the amplitude of the fundamental frequency.

PWM switching for the three-phase inverter is similar to that of the single-phase inverter. Basically, each switch is controlled by comparing a sinusoidal reference wave with a triangular carrier wave. The fundamental frequency of the output is the same as that of the reference wave, and the amplitude of the output is determined by the relative amplitudes of the reference and carrier waves.

As in the case of the six-step three-phase inverter, switches in Fig. 8-28a are controlled in pairs (S_1, S_4), (S_2, S_5), and (S_3, S_6). When one switch in a pair is closed, the other is open. Each pair of switches requires a separate sinusoidal reference

wave. The three reference sinusoids are 120° apart to produce a balanced three-phase output. Figure 8-29a shows a triangular carrier and the three reference waves. Switch controls are such that

S_1 is on when $v_a > v_{tri}$

S_2 is on when $v_c > v_{tri}$

S_3 is on when $v_b > v_{tri}$

S_4 is on when $v_a < v_{tri}$

S_5 is on when $v_c < v_{tri}$

S_6 is on when $v_b < v_{tri}$

Harmonics will be minimized if the carrier frequency is chosen to be an odd triple multiple of the reference frequency, that is, 3, 9, 15, . . . times the reference. Figure 8-29b shows the line-to-line output voltages for a PWM three-phase inverter.

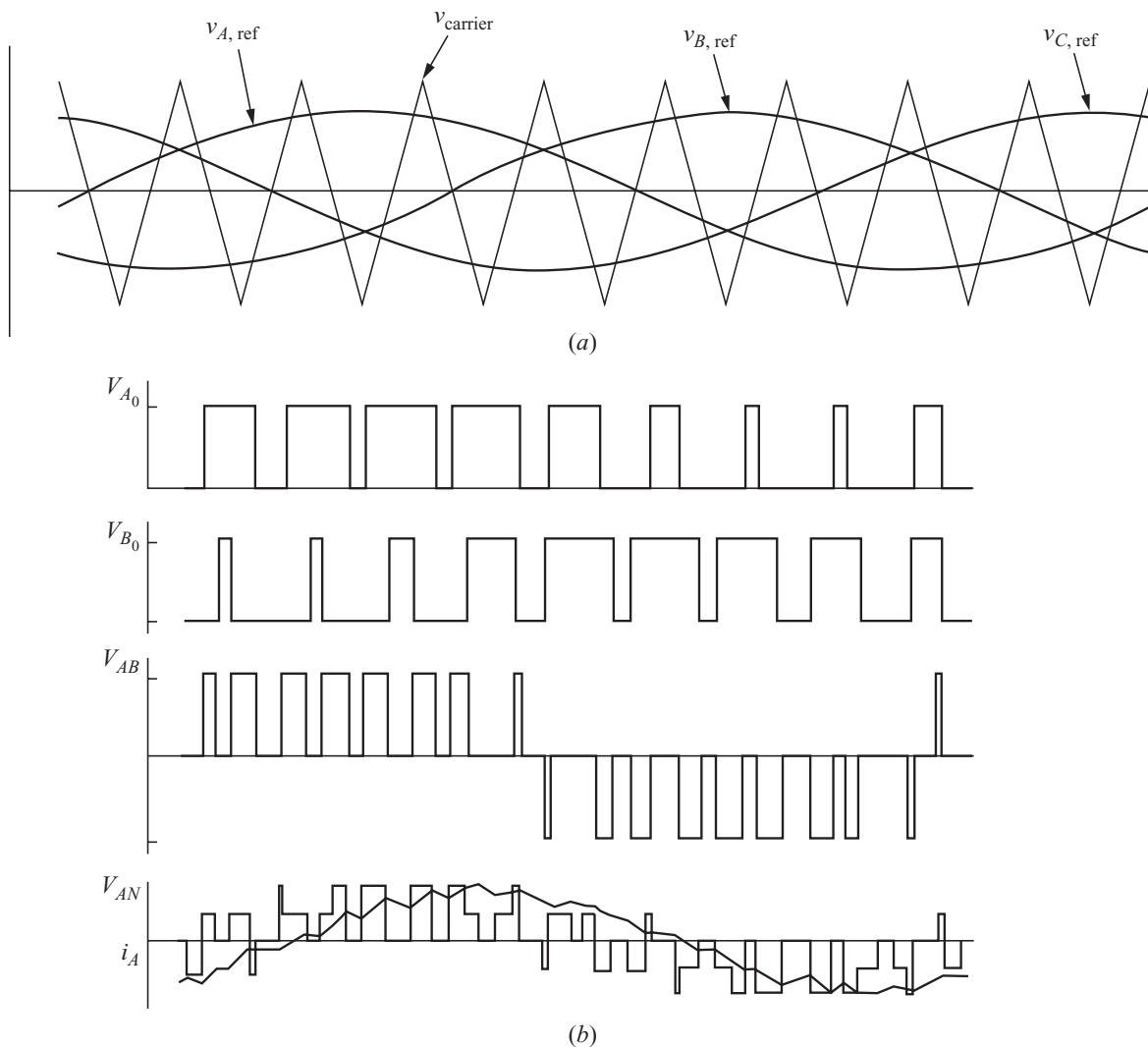


Figure 8-29 (a) Carrier and reference waves for PWM operation with $m_f = 9$ and $m_a = 0.7$ for the three-phase inverter of Fig. 8-28a; (b) Output waveforms—current is for an RL load.

The Fourier coefficients for the line-to-line voltages for the three-phase PWM switching scheme are related to those of single-phase bipolar PWM (V_n in Table 8-3) by

$$V_{n3} = \sqrt{A_{n3}^2 + B_{n3}^2} \quad (8-43)$$

where

$$\begin{aligned} A_{n3} &= V_n \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi}{3}\right) \\ B_{n3} &= V_n \cos\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi}{3}\right) \end{aligned} \quad (8-44)$$

Significant Fourier coefficients are listed in Table 8-8.

Table 8-8 Normalized Amplitudes V_{n3}/V_{dc} for Line-to-Line Three-Phase PWM Voltages

	$m_a=1$	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1
$n = 1$	0.866	0.779	0.693	0.606	0.520	0.433	0.346	0.260	0.173	0.087
$m_f = 2$	0.275	0.232	0.190	0.150	0.114	0.081	0.053	0.030	0.013	0.003
$2m_f = 1$	0.157	0.221	0.272	0.307	0.321	0.313	0.282	0.232	0.165	0.086

Multilevel Three-Phase Inverters

Each of the multilevel inverters described in Sec. 8.9 can be expanded to three-phase applications. Figure 8-30 shows a three-phase diode-clamped multilevel inverter circuit. This circuit can be operated to have a stepped-level output similar to the six-step converter, or, as is most often the case, it can be operated to have a pulse-width-modulated output.

8.16 PSPICE SIMULATION OF THREE-PHASE INVERTERS

Six-Step Three-Phase Inverters

PSpice circuits that will simulate a six-step three-phase inverter are shown in Fig. 8-31. The first circuit is for a complete switching scheme described in Fig. 8-28. Voltage-controlled switches with feedback diodes are used for switching. (The full version of PSpice is required for this circuit.) The second circuit is for generating the appropriate output voltages for the converter so load currents can be analyzed. The output nodes of the inverter are nodes A , B , and C . The parameters shown are those in Example 8-12.

PWM Three-Phase Inverters

The circuit in Fig. 8-32 produces the voltages of the PWM three-phase inverter without showing the switching details. Dependent sources compare sine waves to a triangular carrier wave, as in Example 8-8 for the single-phase case.

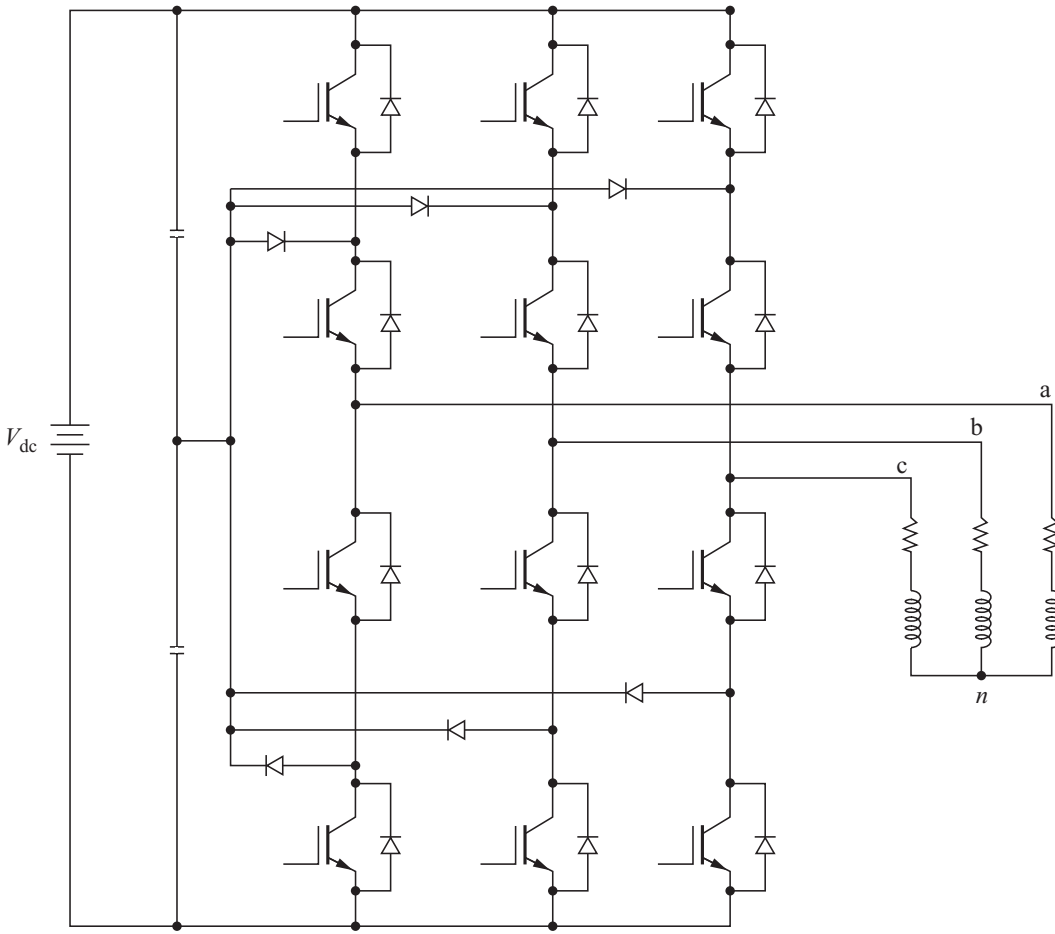


Figure 8-30 A three-phase diode-clamped multilevel inverter.

8.17 INDUCTION MOTOR SPEED CONTROL

The speed of an induction motor can be controlled by adjusting the frequency of the applied voltage. The synchronous speed ω_s of an induction motor is related to the number of poles p and the applied electrical frequency ω by

$$\omega_s = \frac{2\omega}{p} \quad (8-45)$$

Slip s , is defined in terms of the rotor speed ω_r

$$s = \frac{\omega_s - \omega_r}{\omega_s} \quad (8-46)$$

and torque is proportional to slip.

If the applied electrical frequency is changed, the motor speed will change proportionally. However, if the applied voltage is held constant when the frequency is lowered, the magnetic flux in the air gap will increase to the point of

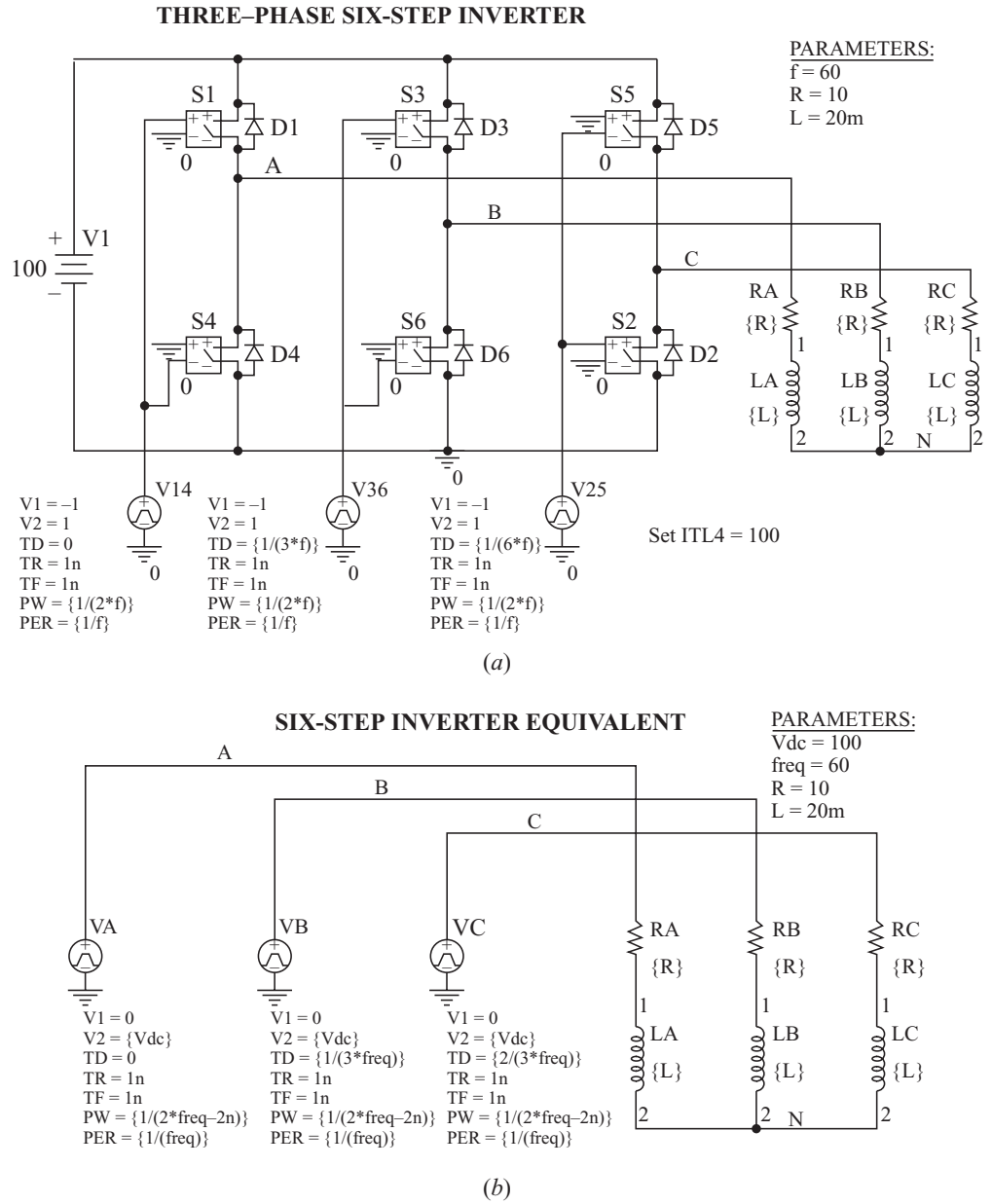


Figure 8-31 (a) A six-step inverter using switches and diodes (requires the full PSpice version); (b) A PSpice circuit for generating three-phase six-step converter voltages.

saturation. It is desirable to keep the air-gap flux constant and equal to its rated value. This is accomplished by varying the applied voltage proportionally with frequency. The ratio of applied voltage to applied frequency should be constant.

$$\frac{V}{f} = \text{constant} \quad (8-47)$$

THREE-PHASE PWM INVERTER

PARAMETERS:

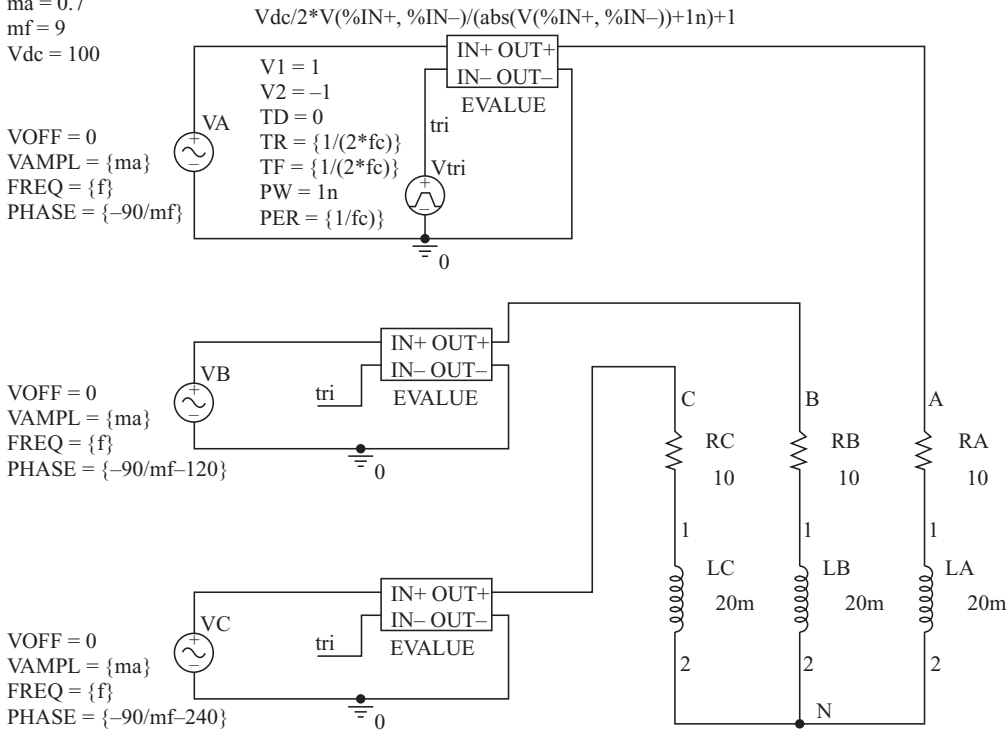
 $f = 60$
$$\mathbf{f_c} = \{\mathbf{f^*mf}\}$$
$$ma = 0.7$$
$$\text{mf} = 9$$
$$V_{dc} = 100$$
$$\text{VOFF} = 0$$
$$\text{VAMPL} = \{\text{ma}\}$$
$$\text{FREQ} = \{f\}$$
$$\text{PHASE} = \{-90/\text{mf}\}$$


Figure 8-32 A PSpice functional circuit for generating three-phase PWM voltages.

The term *volts/hertz control* is often used for this situation. The induction motor torque-speed curves of Fig. 8-33 are for different frequencies and constant volts/hertz.

The six-step inverter can be used for this application if the dc input is adjustable. In the configuration of Fig. 8-34, an adjustable dc voltage is produced from a controlled rectifier, and an inverter produces an ac voltage at the desired

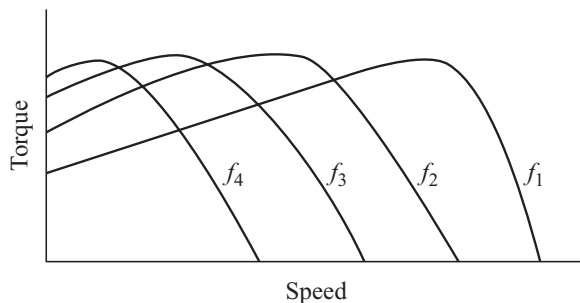


Figure 8-33 Induction motor torque-speed curves for constant volts/hertz variable-speed control.

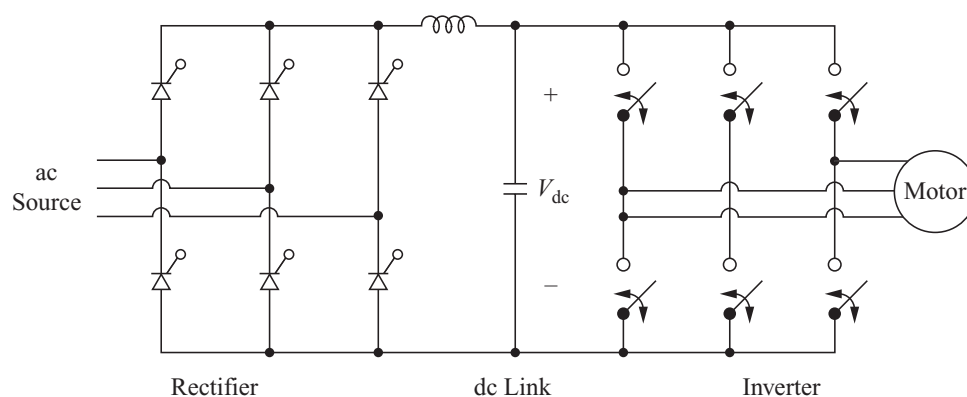


Figure 8-34 AC-AC converter with a dc link.

frequency. If the dc source is not controllable, a dc-dc converter may be inserted between the dc source and the inverter.

The PWM inverter is useful in a constant volts/hertz application because the amplitude of the output voltage can be adjusted by changing the amplitude modulation ratio m_a . The dc input to the inverter can come from an uncontrolled source in this case. The configuration in Fig. 8-34 is classified as an ac-ac converter with a dc link between the two ac voltages.

8.18 Summary

- The full- or half-bridge converters can be used to synthesize an ac output from a dc input.
- A simple switching scheme produces a square wave voltage output, which has a Fourier series that contains the odd harmonic frequencies of amplitudes

$$V_n = \frac{4V_{dc}}{n\pi}$$

- Amplitude and harmonic control can be implemented by allowing a zero-voltage interval of angle α at each end of a pulse, resulting in Fourier coefficients

$$V_n = \left(\frac{4V_{dc}}{n\pi} \right) \cos(n\alpha)$$

- Multilevel inverters use more than one dc voltage source or split a single voltage source with a capacitor voltage divider to produce multiple voltage levels on the output of an inverter.
- Pulse-width modulation (PWM) provides amplitude control of the fundamental output frequency. Although the harmonics have large amplitudes, they occur at high frequencies and are filtered easily.
- Class D audio amplifiers use PWM techniques for high efficiency.
- The six-step inverter is the basic switching scheme for producing a three-phase ac output from a dc source.
- A PWM switching scheme can be used with a three-phase inverter to reduce the THD of the load current with modest filtering.
- Speed control of induction motors is a primary application of three-phase inverters.

8.19 Bibliography

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Problems

Square-Wave Inverter

- 8-1.** The square-wave inverter of Fig. 8-1a has $V_{dc} = 125$ V, an output frequency of 60 Hz, and a resistive load of $12.5\ \Omega$. Sketch the currents in the load, each switch, and the source, and determine the average and rms values of each.
- 8-2.** A square-wave inverter has a dc source of 96 V and an output frequency of 60 Hz. The load is a series RL load with $R = 5\ \Omega$ and $L = 100$ mH. When the load is first energized, a transient precedes the steady-state waveform described

- by Eq. (8-5). (a) Determine the peak value of the steady-state current. (b) Using Eq. (8-1) and assuming zero initial inductor current, determine the maximum current that occurs during the transient. (c) Simulate the circuit with the PSpice input file of Fig. 8.4a and compare the results with parts (a) and (b). How many periods must elapse before the current reaches steady state? How many L/R time constants elapse before steady state?
- 8-3.** The square-wave inverter of Fig. 8-3 has a dc input of 150 V and supplies a series RL load with $R = 20\ \Omega$ and $L = 40\text{ mH}$. (a) Determine an expression for steady-state load current. (b) Sketch the load current and indicate the time intervals when each switch component (Q1, D1; . . . Q4, D4) is conducting. (c) Determine the peak current in each switch component. (d) What is the maximum voltage across each switch? Assume ideal components.
- 8-4.** A square-wave inverter has a dc source of 125 V, an output frequency of 60 Hz, and an RL series load with $R = 20\ \Omega$ and $L = 25\text{ mH}$. Determine (a) an expression for load current, (b) rms load current, and (c) average source current.
- 8-5.** A square-wave inverter has an RL load with $R = 15\ \Omega$ and $L = 10\text{ mH}$. The inverter output frequency is 400 Hz. (a) Determine the value of the dc source required to establish a load current that has a fundamental frequency component of 8 A rms. (b) Determine the THD of the load current.
- 8-6.** A square-wave inverter supplies an RL series load with $R = 25\ \Omega$ and $L = 25\text{ mH}$. The output frequency is 120 Hz. (a) Specify the dc source voltage such that the load current at the fundamental frequency is 2.0 A rms. (b) Verify your results with PSpice. Determine the THD from PSpice.
- 8-7.** A square-wave inverter has a dc input of 100 V, an output frequency of 60 Hz, and a series RLC combination with $R = 10\ \Omega$, $L = 25\text{ mH}$, and $C = 100\ \mu\text{F}$. Use the PSpice simplified square-wave inverter circuit of Fig. 8-4a to determine the peak and rms value of the steady-state current. Determine the total harmonic distortion of the load current. On a printout of one period of the current, indicate the intervals where each switch component in the inverter circuit of Fig. 8-3 is conducting for this load if that circuit were used to implement the converter.

Amplitude and Harmonic Control

- 8-8.** For the full-bridge inverter, the dc source is 125 V, the load is a series RL connection with $R = 10\ \Omega$ and $L = 20\text{ mH}$, and the switching frequency is 60 Hz. (a) Use the switching scheme of Fig. 8-5 and determine the value of α to produce an output with an amplitude of 90 V at the fundamental frequency. (b) Determine the THD of the load current.
- 8-9.** An inverter that produces the type of output shown in Fig. 8-5a is used to supply an RL series load with $R = 10\ \Omega$ and $L = 35\text{ mH}$. The dc input voltage is 200 V and the output frequency is 60 Hz. (a) Determine the rms value of the fundamental frequency of the load current when $\alpha = 0$. (b) If the output fundamental frequency is lowered to 30 Hz, determine the value of α required to keep the rms current at the fundamental frequency at the same value of part (a).
- 8-10.** Use the PSpice circuit of Fig. 8-7a to verify that (a) the waveform of Fig. 8-5a with $\alpha = 30^\circ$ contains no third harmonic frequency and (b) the waveform of Fig. 8-5a with $\alpha = 18^\circ$ contains no fifth harmonic.
- 8-11.** (a) Determine the value of α that will eliminate the seventh harmonic from the inverter output of Fig. 8-5a. (b) Verify your answer with a PSpice simulation.

- 8-12.** Determine the rms value of the notched waveform to eliminate the third and fifth harmonics in Fig. 8-6.
- 8-13.** Use PSpice to verify that the notched waveform of Fig. 8-6c contains no third or fifth harmonic. What are the magnitudes of the fundamental frequency and the first four nonzero harmonics? (The piecewise linear type of source may be useful.)

Multilevel Inverters

- 8-14.** For a multilevel inverter having three separate dc sources of 48 V each, $\alpha_1 = 15^\circ$, $\alpha_2 = 25^\circ$, and $\alpha_3 = 55^\circ$. (a) Sketch the output voltage waveform. (b) Determine the Fourier coefficients through $n = 9$. (c) Determine the modulation index M_i .
- 8-15.** For a three-source multilevel inverter, select values of α_1 , α_2 , and α_3 such that the third harmonic frequency ($n = 3$) in the output voltage waveform is eliminated. Determine the modulation index M_i for your selection.
- 8-16.** The five-source multilevel inverter of Fig. 8-11 has $\alpha_1 = 16.73^\circ$, $\alpha_2 = 26.64^\circ$, $\alpha_3 = 46.00^\circ$, $\alpha_4 = 60.69^\circ$, and $\alpha_5 = 62.69^\circ$. Determine which harmonics will be eliminated from the output voltage. Determine the amplitude of the fundamental-frequency output voltage.
- 8-17.** The concept of the two-source multilevel inverters of Figs. 8-9 and 8-11 is extended to have three independent sources and H bridges and three delay angles α_1 , α_2 , and α_3 . Sketch the voltages at the output of each bridge of a three-source multilevel converter such that the average power from each source is the same.

Pulse-Width-Modulated Inverters

- 8-18.** The dc source supplying an inverter with a bipolar PWM output is 96 V. The load is an RL series combination with $R = 32 \Omega$ and $L = 24$ mH. The output has a fundamental frequency of 60 Hz. (a) Specify the amplitude modulation ratio to provide a 54-V rms fundamental frequency output. (b) If the frequency modulation ratio is 17, determine the total harmonic distortion of the load current.
- 8-19.** The dc source supplying an inverter with a bipolar PWM output is 250 V. The load is an RL series combination with $R = 20 \Omega$ and $L = 50$ mH. The output has a fundamental frequency of 60 Hz. (a) Specify the amplitude modulation ratio to provide a 160-V rms fundamental frequency output. (b) If the frequency modulation ratio is 31, determine the total harmonic distortion of the load current.
- 8-20.** Use PSpice to verify that the design in Example 8-9 meets the THD specifications.
- 8-21.** Design an inverter that has a PWM output across an RL series load with $R = 10 \Omega$ and $L = 20$ mH. The fundamental frequency of the output voltage must be 120 V rms at 60 Hz, and the total harmonic distortion of the load current must be less than 8 percent. Specify the dc input voltage, the amplitude modulation ratio m_a , and the switching frequency (carrier frequency). Verify the validity of your design with a PSpice simulation.
- 8-22.** Design an inverter that has a PWM output across an RL series load with $R = 30 \Omega$ and $L = 25$ mH. The fundamental frequency of the output voltage must be 100 V rms at 60 Hz, and the total harmonic distortion of the load current must be less than 10 percent. Specify the dc input voltage, the amplitude modulation ratio m_a , and the switching frequency (carrier frequency). Verify the validity of your design with a PSpice simulation.

- 8-23.** Pulse-width modulation is used to provide a 60-Hz voltage across a series RL load with $R = 12\ \Omega$ and $L = 20\text{ mH}$. The dc supply voltage is 150 V. The amplitude of the 60-Hz voltage is to be 120 V. Use PSpice to obtain the current waveform in the load and the THD of the current waveform in the load. Use (a) bipolar PWM with $m_f = 21$, (b) bipolar PWM with $m_f = 41$, and (c) unipolar PWM with $m_f = 10$.

Three-Phase Inverters

- 8-24.** A six-step three-phase inverter has a 250-V dc source and an output frequency of 60 Hz. A balanced Y-connected load consists of a series 25- Ω resistance and 20-mH inductance in each phase. Determine (a) the rms value of the 60-Hz component of load current and (b) the THD of the load current.
- 8-25.** A six-step three-phase inverter has a 400-V dc source and an output frequency that varies from 25 to 100 Hz. The load is a Y connection with a series 10- Ω resistance and 30-mH inductance in each phase. (a) Determine the range of the rms value of the fundamental-frequency component of load current as the frequency is varied. (b) What is the effect of varying frequency on the THD of the load current and the THD of the line-to-neutral voltage?
- 8-26.** A six-step three-phase inverter has an adjustable dc input. The load is a balanced Y connection with a series RL combination in each phase, with $R = 5\ \Omega$ and $L = 50\text{ mH}$. The output frequency is to be varied between 30 and 60 Hz. (a) Determine the range of the dc input voltage required to maintain the fundamental-frequency component of current at 10 A rms. (b) Use PSpice to determine the THD of load current in each case. Determine the peak current and rms load current for each case.